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Daniel Vieira

ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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ROBUST CONTROL OF LINEAR SYSTEMS WITH SWITCHED ACTUATORS SUBJECTED TO DWELL-TIME CONSTRAINTS

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"I am and always will be the optimist.
The hoper of far-flung hopes, and the dreamer of improbable dreams."
— The Eleventh Doctor

Resumo

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.

Abstract

Switched actuators are difficult to handle because of some characteristic constraints of their nature. but they are efficient in numerous engineering applications, being applied, for example, in attitude control systems for space vehicles. Thus, knowing how to use this type of actuator is important and necessary. The main problem to be solved in this thesis is the design of a controller that guarantees the robustness of a system with switched actuators subjected to time constraints that limit the response window of their commands. In this approach, a limit cycle target trajectory search is performed and then the error of the states compared to the target trajectory is calculated. Then a predictive control technique is applied to correct the error and get the system to the cyclic target trajectory

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List of Abbreviations and Acronyms

CTq	computed torque
DC	direct current
EAR	Equação Algébrica de Riccati
GDL	graus de liberdade
ISR	interrupção de serviço e rotina
LMI	linear matrices inequalities
MIMO	multiple input multiple output
PD	proporcional derivativo
PID	proporcional integrativo derivativo
PTP	point to point
UARMII	Underactuated Robot Manipulator II
VSC	variable structure control

List of Symbols

a	Distância
$\mathbf{a}$	Vetor de distâncias
$\mathbf{e}_j$	Vetor unitário de dimensão n e com o j-ésimo componente igual a 1
$\mathbf{K}$	Matriz de rigidez
m_1	Massa do cumpim
δ_{k-k_f}	Delta de Kronecker no instante k_f

Contents

1 Introduction

1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$. Where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) \in \{1, 2, ..., M\}$ is the switching state, which is associated to the matrix $A_{\sigma(t)} \in \mathbb{R}^{n\times n}$ and the vector $b_{\sigma(t)} \in \mathbb{R}^n$ at each time t.

The use of switched affine systems to solve problems in engineering can be seen in fields like biochemical reactions, where Parise *et al.* (2018) uses switched affine systems to represent the moments of evolution in a generic biochemical reaction network; urban traffic, where Hajiahmadi *et al.* (2016) approaches the urban traffic control; power converters, where Benmiloud *et al.* (2019) also uses this approach and other areas where the behavior of the system can be described as a switched affine system.

There is extensive research in this class of controllers, and most of them solve the problem by driving the system states to the neighborhood of the target states, not considering the trajectory behavior of the system. This can reduce the performance of the system when the system is affected by the trajectory of the limit cycle in steady operation.

This problem has been the target of many studies, Benmiloud *et al.* (2019) uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired limit cycle and uses as an example a DC-DC converter. by Egidio *et al.* (2020b), which proposes a control design of a state-dependent switching function for discrete-time switched affine systems to drive the system to the desired limit cycle that is defined by design associated with the expected performance of the system. The solution was applied in an academic numerical example and a theoretical DC-DC converter. In Egidio *et al.* (2020a) the author expanded the work to continuous-time switched affine systems.

Some studies form the basis for the present work:

Vieira (2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

1.2 Contributions of the present work

This present work proposes to expand the solution from Marcolino (2018) and provide a broader solution that can be used for systems that have their dynamics changing over time.

In a system that can be represented by a switched affine system with the form

$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}$

there are cases in which the dynamic response of the system can change depending on the operation mode. This can be represented in switched affine systems with proper matrices $A_{\sigma(t)}$ and $b_{\sigma(t)}$ for each operation mode. With the current implementation of Marcolino (2018) considering an invariant system during its operation, there is a need for expanding the solution so it can represent more systems and solve more engineering problems that can be expressed as switched affine systems.

1.3 Outline of the remaining chapters

The remaining of this text is organized as follows. Chapter 2 describes the problem of finding a target periodic trajectory for the given switched affine system. Chapter 3 proposes then a predictive control law that drives the system to the target cyclic trajectory. Finally, chapter 4 brings some remarks and considerations of the work.

2 Determination of Target Periodic Trajectories

The goal is to drive a system defined by an affine dynamic equation (eq. 2.1) to an optimal periodic trajectory, so the first step is to generate this trajectory. This trajectory is submitted to restrictions imposed by the dynamic of the system and also switching time restrictions.

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \tag{2.1}$$

Following the approach from Patino *et al.* (2010) we must establish a trajectory that yields the optimal time sequence τ^∞, the modes sequence Ω^∞, and its length $s^\infty(1 < s^\infty < s_{max})$. To reduce the oscillations around a desired operation point, we choose an integral quadratic criterion centered on the average reference value $\bar{x}^\infty$:

$$J(\tau^\infty, \Omega^\infty, s^\infty) = \min_{\tau,\Omega,s} \int_0^{T_p} \Big[x(t) - \bar{x}^\infty\Big]^T Q \Big[x(t) - \bar{x}^\infty\Big] dt \tag{2.2}$$

where $Q = Q^T > 0$ and T_p is the period of the cycle.

Since the trajectory is cyclic we can impose the constraint:

$$x(0) = x(T_p) \tag{2.3}$$

2.1 Optimization

2.1.1 System Dynamics and Sequence

To begin, we create a list where each of its elements signifies the dynamics of the system. This is exemplified by the matrices A, B, C, and D. The order in which the elements appear in this list is defined by the sequence of modes. In simpler terms, the positioning of these elements signifies the order they will be executed.

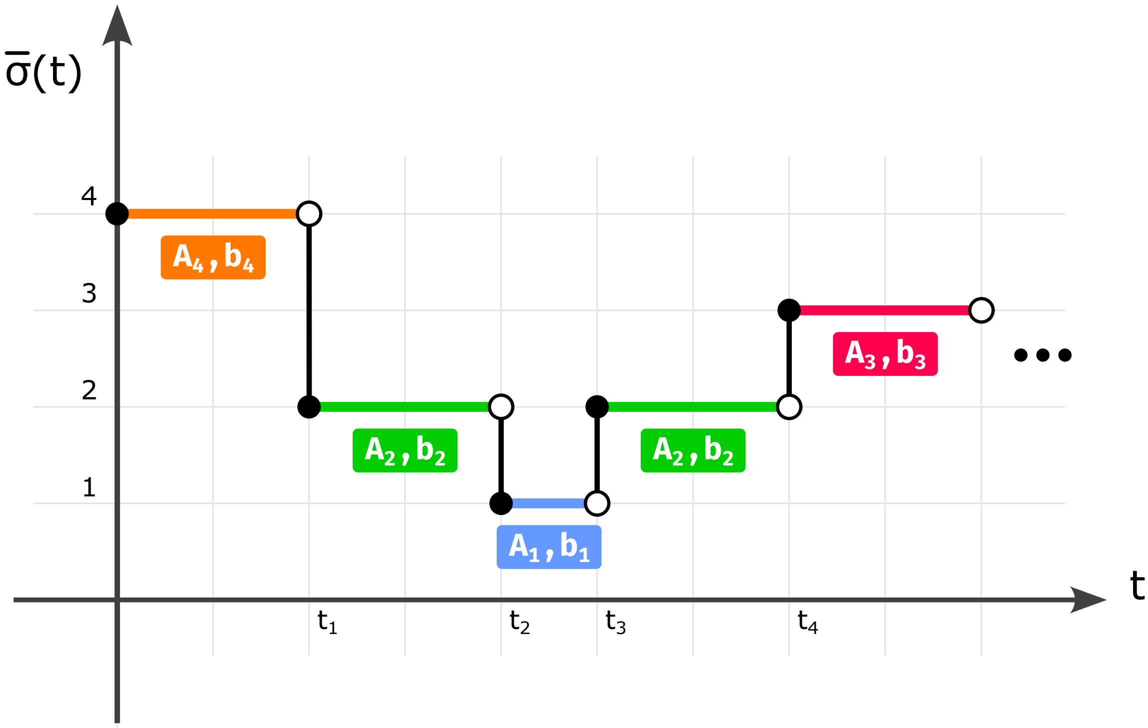

FIGURE 2.1 – Change in dynamics over trajectory time

In Figure 2.1, we see a trajectory example. For this illustration, the system's list of dynamics is $\Big\{[A_1, b_1], [A_2, b_2], [A_3, b_3], [A_4, b_4]\Big\}$, and the sequence of modes is $[4, 2, 1, 2, 3]$. It's worth noting that a particular system dynamic can be employed multiple times or might not be used at all, depending on the desired target trajectory. The sequence of modes and the maximum period for the cycle are predetermined for this problem and the goal is to achieve the best trajectory that minimizes the cost function.

2.1.2 Constraints Definition

Once the list and sequence have been established, we need to apply the constraints. These constraints dictate the cycle's maximum duration, as well as the minimum and the maximum time permitted between switches in modes.

2.1.3 Optimization with FMINCON

For optimization purposes, the function FMINCON is employed. This function works in conjunction with a cost function. The cost function is designed to accept switching times generated from the optimization process. Its primary goal is to evaluate and discern the most cost-effective result that respects the imposed constraints.

FMINCON is a method of optimization in MATLAB's libraries that implements the method SQP (Sequential Quadratic Programming). The basic SQP algorithm is described in Chapter 18 of Nocedal e Wright (2006).

2.1.4 Cost Function and Trajectory Simulation

Taking a closer look at the cost function, its role is to simulate the trajectory using the dynamics taken from the previously defined list, all in line with the mode sequence. This trajectory is cyclic and to enforce this cyclic nature, the initial state's condition is calculated in such a way that the end product is a cyclic trajectory.

2.1.5 Cost Computation

The method of determining the cost is by computing the integral of the trajectory, centered around the average reference value, denoted as $\bar{x}^{\infty}$. The result is compliant with the system's dynamic constraints and also matches the properties of the final trajectory.

2.2 Calculating x_0

We need to compute the initial condition for the system for each evaluation of the cost function so that we have a periodic trajectory, meaning that the last states of the trajectory we the same as the initial condition.

We can have different dynamics for each switching instant, so the system behaves as

$$\begin{aligned}
[0 \to t_1] &: \dot{x} = A_{\sigma(0)}x + b_{\sigma(0)} \\
[t_1 \to t_2] &: \dot{x} = A_{\sigma(t_1)}x + b_{\sigma(t_1)} \\
&\vdots \\
[t_{N-1} \to t_N] &: \dot{x} = A_{\sigma(t_{N-1})}x + b_{\sigma(t_{N-1})}
\end{aligned} \tag{2.4}$$

We can rewrite the solution for the first region as

$$\begin{aligned}
x(t_1) &= e^{A_{\sigma(0)}t_1}x(0) + \int_0^{t_1} e^{A_{\sigma(0)}(t_1-\tau)}b_{\sigma(0)}d\tau \\
x(t_1) &= e^{A_{\sigma(0)}t_1}x(0) + \int_0^{t_1} e^{A_{\sigma(0)}(t_1-\tau)}b_{\sigma(0)}d\tau \\
x(t_1) &= F_0 x(0) + g_0
\end{aligned} \tag{2.5}$$

then the behavior of the other regions of the system is

$$\begin{aligned} x(t_1) &= F_0 x(0) + g_0 \\ x(t_2) &= F_1 x(1) + g_1 \\ &\vdots \\ x(t_N) &= F_{N-1} x(N-1) + g_{N-1} \end{aligned} \tag{2.6}$$

with

$$F_i = e^{A_{\sigma(t_i)}(t_{i+1}-t_i)} \tag{2.7}$$

$$g_i = \int_{t_i}^{t_{i+1}} e^{A_{\sigma(t_i)}(t_{i+1}-\tau)} b_{\sigma(t_i)} d\tau = \int_0^{t_{i+1}-t_i} e^{A_{\sigma(t_i)}\tau} b_{\sigma(t_i)} d\tau \tag{2.8}$$

From (2.6) we can establish the relation between $x(t_N)$ and $x(0)$

$$\begin{aligned} x(t_N) = {} & F_{N-1}F_{N-2}\dots F_0 x(0) + \\ & F_{N-1}F_{N-2}\dots F_1 g_0 + F_{N-1}F_{N-2}\dots F_2 g_1 + \dots + F_{N-1} g_{N-2} + g_{N-1} \\ x(t_N) = {} & F_N F_{N-1}\dots F_1 x(0) + c \end{aligned} \tag{2.9}$$

We know that $x(t_N) = x(0)$, so the solution for $x(0)$ is

$$\begin{aligned} x(0) &= F_{N-1}F_{N-2}\dots F_0 x(0) + c \\ \Big(I - F_{N-1}F_{N-2}\dots F_0\Big) x(0) &= c \\ x(0) &= \Big(I - F_{N-1}F_{N-2}\dots F_0\Big)^{-1} c \end{aligned} \tag{2.10}$$

where

$$c = F_{N-1}F_{N-2}\dots F_1 g_0 + F_{N-1}F_{N-2}\dots F_2 g_1 + \dots + F_{N-1} g_{N-2} + g_{N-1} \tag{2.11}$$

assuming that the inverse indicated in (2.10) exists.

2.3 Application Examples

2.3.1 Buck-boost converter

Considering a Buck-Boost converter in a Continuous Conduction Mode (CCM) whose topology is shown in Figure 2.2.

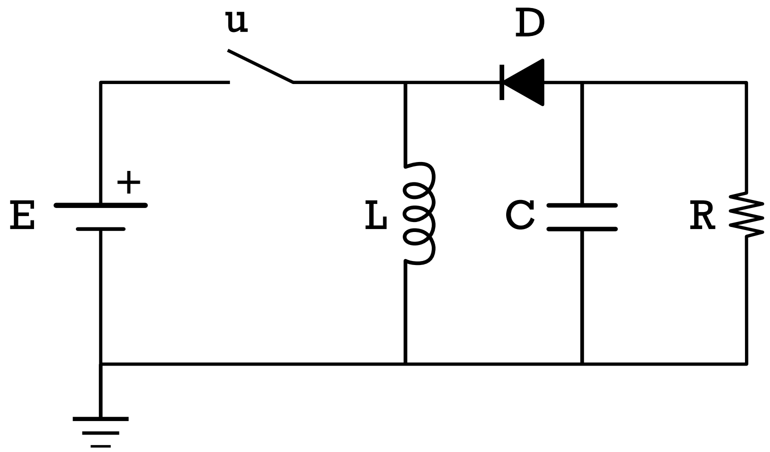

FIGURE 2.2 – Buck-Boost converter

The state variables are the capacitor voltage v_c and the inductor current i_L. Its dynamics are given by the equation:

$$\dot{x} = \begin{cases} A_1 x + b_1, \text{ for u} = 1 \text{ (closed switch - Mode 1)} \\ A_2 x + b_2, \text{ for u} = 0 \text{ (open switch - Mode 2)} \end{cases} \tag{2.12}$$

where

$$\begin{aligned} A_1 &= \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & \frac{1}{L_1} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \\ b_1 &= \begin{bmatrix} \frac{E}{L} & 0 \end{bmatrix}^T \qquad b_2 = \begin{bmatrix} 0 & 0 \end{bmatrix}^T \end{aligned} \tag{2.13}$$

where $R = L = C = E = 1$.

The list of dynamics for this example is

$$\Big\{ \big[A_1, b_1\big], \big[A_2, b_2\big] \Big\} \tag{2.14}$$

where

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tag{2.15}$$

$$A_2 = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.16}$$

The criteria for the optimization is to compute the trajectory to minimize oscillation around the average value $\big[2, -1\big]$ with the weight matrix $Q = diag\big[1, 1\big]$, the maximum cycle period of $T_{p,max} = 1s$ and minimum dwell time of $t_{min} = 0.25s$.

The cost function takes a vector representing the dt for each switching moment as its input. For instance, with two switching instances, the vector could be $\big[0.25, 0.35\big]$. This implies the signal begins at time 0.0, transitions to the next dynamic at $0.0 + 0.25 = 0.25$, and then switches again at $0.25 + 0.35 = 0.6$. This gives us the signal sequence $\big[0.0, 0.25, 0.6\big]$.

In the Buck-Boost converter example, the initial values for the switching vector are represented by $[0.25, 0.25]$. The lower bounds are defined as $[0.25, 0.25]$ and the upper bounds are set to $[1.5, 1.5]$. The solution provided by the FMINCON optimization yields $\tau^\infty = \{0, 0.2509, 0.5\}$ and $\Omega^\infty = \{1, 2\}$.

The outcome under this condition is displayed in Figure (2.2), showcasing the cyclic trajectory around the average value $[2, -1]$. The corresponding command signal can be seen in Figure (2.4).

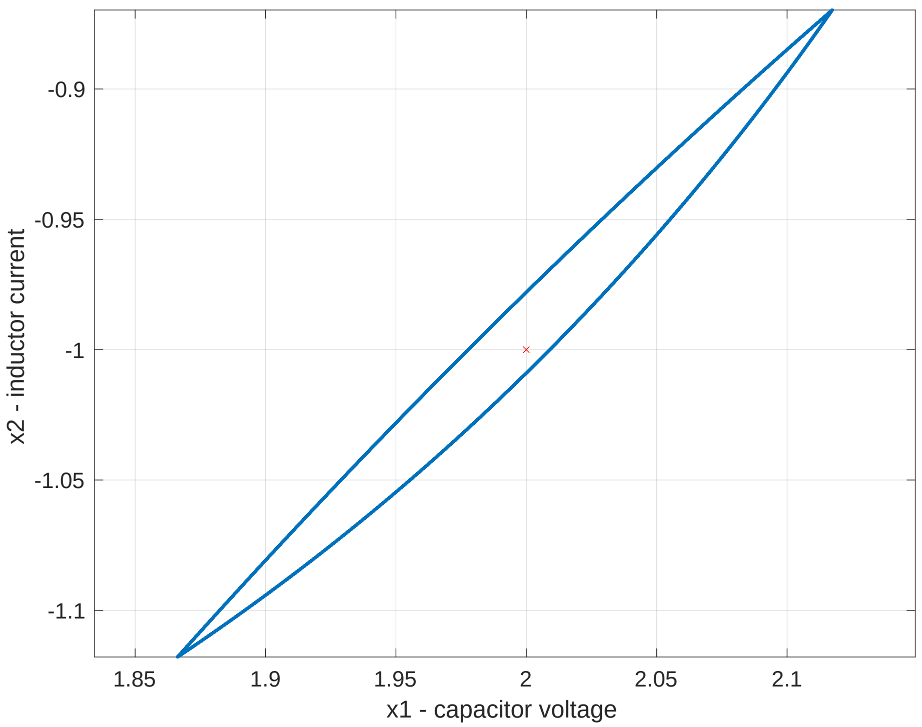

FIGURE 2.3 – Buck-Boost converter: State trajectory

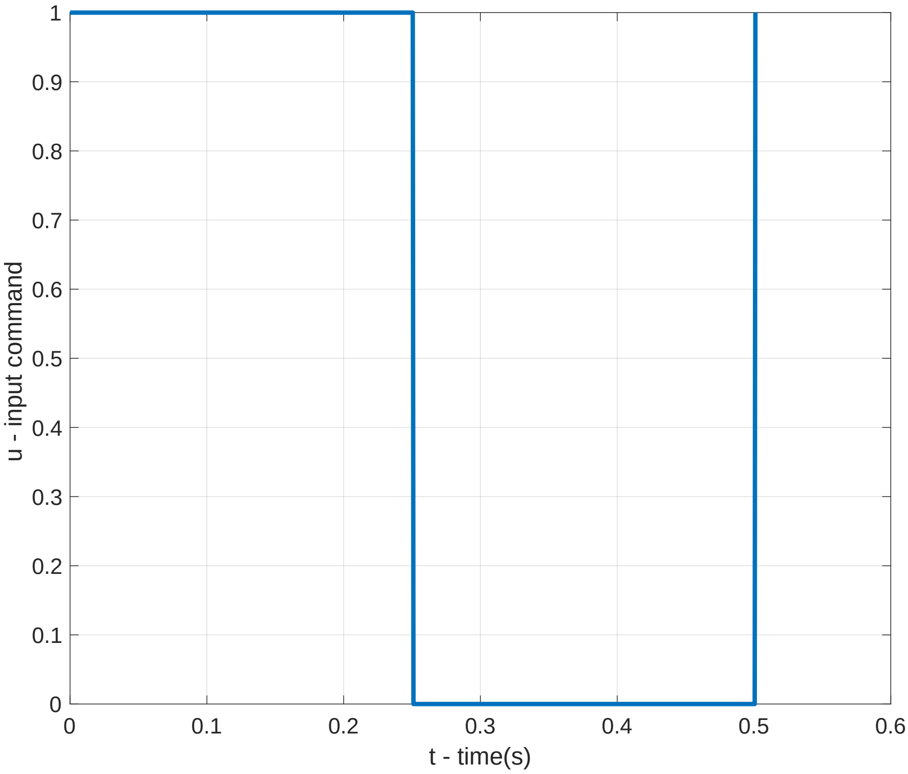

FIGURE 2.4 – Buck-Boost converter: Command signal

2.3.2 Multilevel converter

Considering a Multilevel converter whose topology is shown in Figure 2.5.

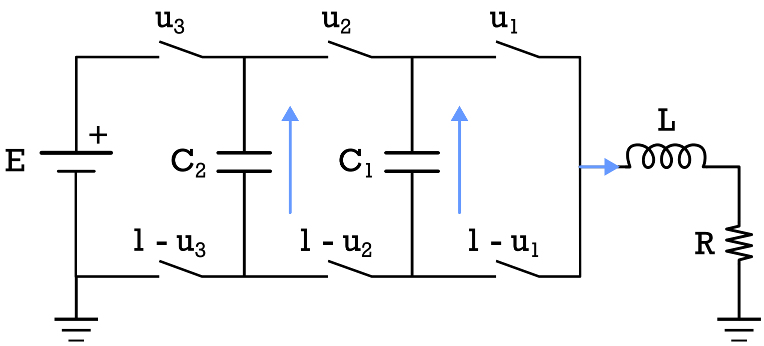

FIGURE 2.5 – Four-level three-cell converter

The state variables are the voltage on the capacitors $x(t) = [v_{c_1}(t), v_{c_2}(t), i_L(t)]$. Where v_{c_1}, v_{c_2} are the voltages in each capacitor and i_L is the load current.

The dynamic of this system can be described as

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} -\frac{x_3(t)}{C_1} & \frac{x_3(t)}{C_1} & 0 \\ 0 & -\frac{x_3(t)}{C_2} & \frac{x_3(t)}{C_2} \\ \frac{x_1(t)}{L} & \frac{x_2(t)-x_1(t)}{L} & \frac{E-x_2(t)}{L} \end{bmatrix} \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}x_3(t) \end{bmatrix} \tag{2.17}$$

rearranging the terms we have

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{u_2-u_1}{C1} \\ 0 & 0 & \frac{u_3-u_2}{C2} \\ \frac{u_1-u_2}{L} & \frac{u_2-u_3}{L} & \frac{-R}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\frac{R}{L}u_3 \end{bmatrix} \tag{2.18}$$

Each combination of the switches determines a different behavior

$$\mathbf{u_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, A_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, b_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.19}$$

$$\mathbf{u_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{1}{C_1} \\ 0 & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.20}$$

$$\mathbf{u_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, A_3 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & \frac{-1}{C_2} \\ \frac{-1}{L} & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, b_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.21}$$

$$\mathbf{u_4} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, A_4 = \begin{bmatrix} 0 & 0 & \frac{1}{C_1} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, b_4 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.22}$$

$$\mathbf{u_5} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, A_5 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & 0 \\ \frac{1}{L} & 0 & \frac{-R}{L} \end{bmatrix}, b_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.23}$$

$$\mathbf{u_6} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, A_6 = \begin{bmatrix} 0 & 0 & \frac{-1}{C_1} \\ 0 & 0 & \frac{1}{C_2} \\ \frac{1}{L} & \frac{-1}{L} & \frac{-R}{L} \end{bmatrix}, b_6 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.24}$$

$$\mathbf{u_7} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, A_7 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{C_1} \\ 0 & \frac{1}{L} & \frac{-R}{L} \end{bmatrix}, b_7 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{2.25}$$

$$\mathbf{u_8} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, A_8 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{-R}{L} \end{bmatrix}, b_8 = \begin{bmatrix} 0 \\ 0 \\ \frac{E}{L} \end{bmatrix} \tag{2.26}$$

With the detailed dynamic for each switching positions we can define the list of elements for the multilevel converter as $\Big\{A_1, A_2, A_3, A_4, A_5, A_6, A_7\Big\}$ and $\Big\{b_1, b_2, b_3, b_4, b_5, b_6, b_7\Big\}$ with

The criteria for the optimization is to compute the trajectory around the average value $\Big[\frac{2}{3}E, \frac{1}{3}E, 1\Big]$ with $E = 30V$. The maximum cycle period is $T_{p,max} = 0.4ms$ and minimum tweel time of $t_{min} = 0.022ms$.

The simulation parameters are $C_1 = 40\mu F$, $C_2 = 40\mu F$, $L = 10mH$ and $R = 10\Omega$ with the weight matrix $Q = diag\Big[10, 5, 20000\Big]$.

Using the optimal switching sequence

$$\Omega^{\infty} = \Big\{1, 2, 4, 8, 3, 1, 5, 8, 5\Big\} \tag{2.27}$$

The optimal switching instant is

$$\tau^{\infty} = \Big\{0, 0.088, 0.110, 0.132, 0.154, 0.176, 0.198, 0.220, 0.242, 0.264\Big\} \tag{2.28}$$

with the initial condition $x(0) = \Big[21.71, 3.29, 1.08\Big]^T$.

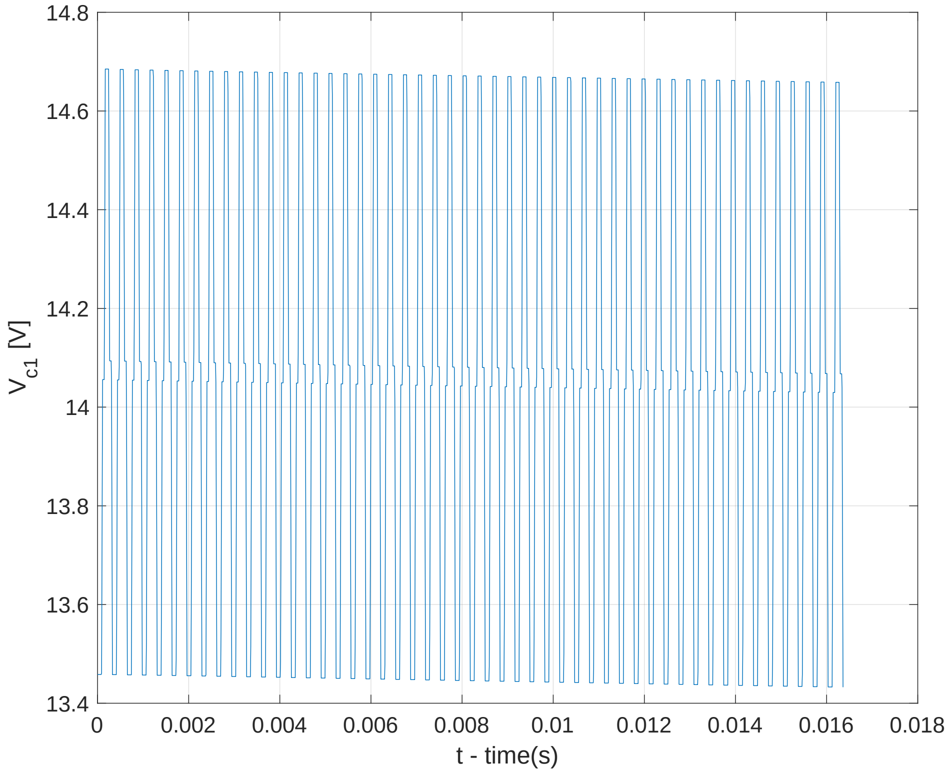

FIGURE 2.6 – Variant Dynamic Circuit 2: Voltage C_1

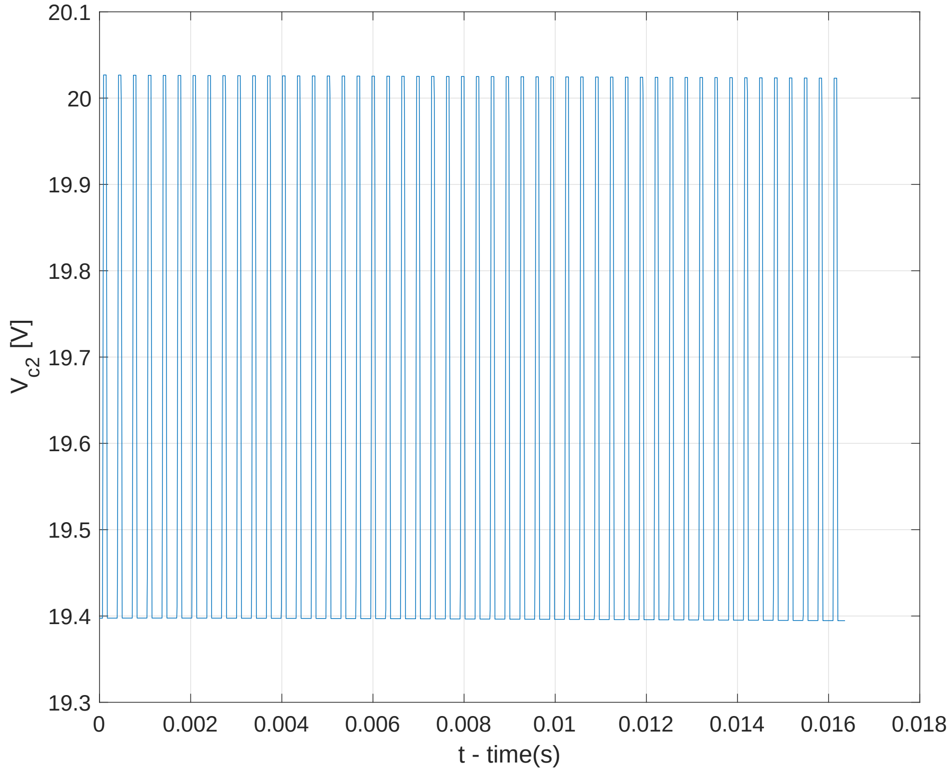

FIGURE 2.7 – Variant Dynamic Circuit 2: Voltage C_2

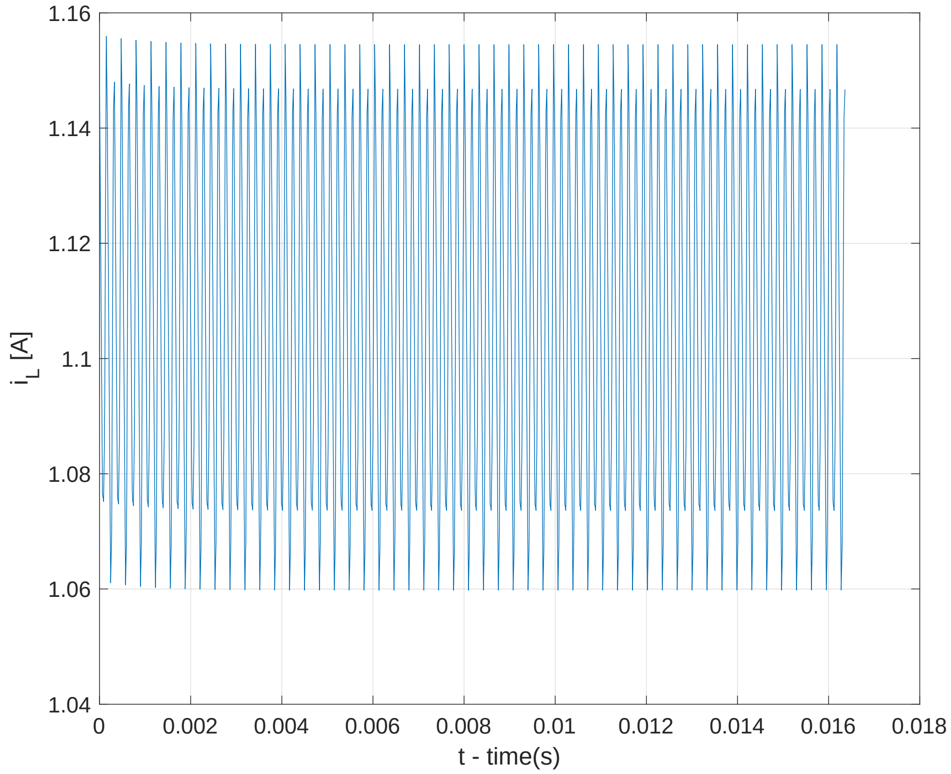

FIGURE 2.8 – Variant Dynamic Circuit 2: Current L

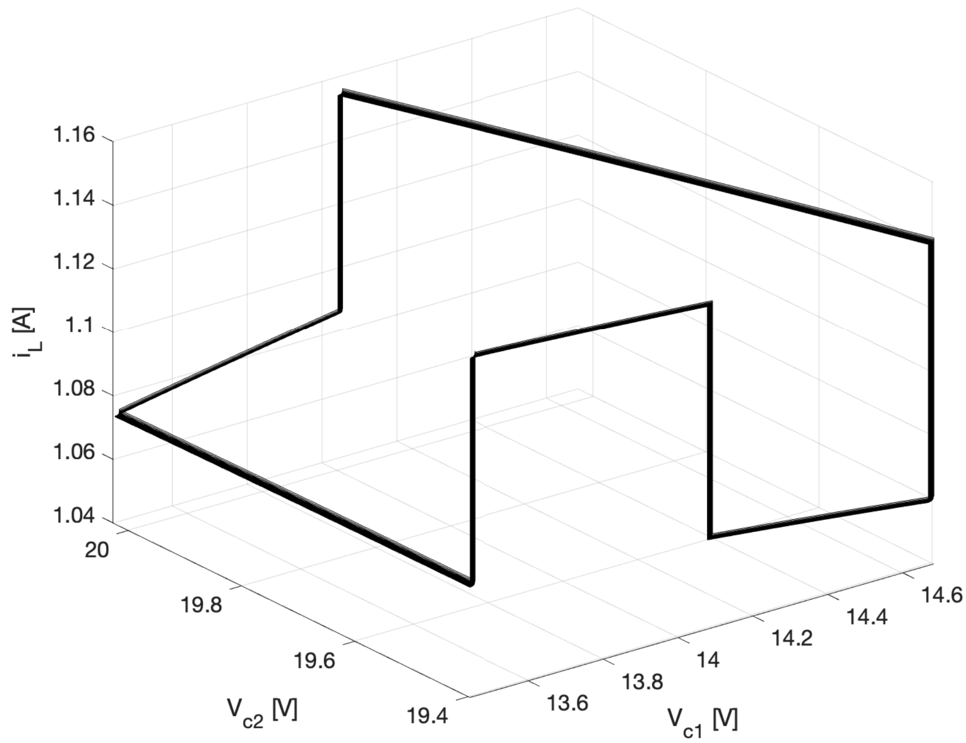

FIGURE 2.9 – Variant Dynamic Circuit 2: State Trajectory

3 Predictive control: Preliminaries and notation

Consider a system with dynamics described by

$$x[k+1] = Ax[k] + Bu[k] \tag{3.1}$$

The objective is to drive the state $x[k]$ to the origin from a given initial condition while respecting state and control constraints in the form

$$x[k] \in \mathcal{X} \tag{3.2}$$

$$u[k] \in \mathcal{U} \tag{3.3}$$

$$\mathcal{U} = \{u \in \mathbb{R}^p : S_u u \leq b_u\} \tag{3.4}$$

and

$$\mathcal{X} = \{x \in \mathbb{R}^n : S_x x \leq b_x\} \tag{3.5}$$

3.1 Cost Function

The solution, if it exists, will not be unique. Therefore, it is necessary to adopt a criterion that allows for choosing the best sequence of control actions to perform the task.

The control problem consists of bringing the states to the origin, so we will employ a cost in the form

$$J = \left\|\hat{x}[k+N|k]\right\|_{P_f}^2 + \sum_{i=0}^{N-1}\left[\left\|\hat{x}[k+i|k]\right\|_Q^2 + \left\|\hat{u}[k+i|k]\right\|_R^2\right] \tag{3.6}$$

and initial condition

$$\hat{x}[k|k] = x[k] \tag{3.7}$$

The expression for J consists of a terminal cost term and a trajectory cost term.

With P_f, Q, R being symmetric and positive-definite, with P_f chosen so that the terminal cost term $\left\|\hat{x}(k+N|k) - \bar{x}\right\|_{P_f}^2$ corresponds to the value function of the infinite horizon Discrete-Time Linear Quadratic Regulator (DLQR).

The cost function can be rewritten using the notation:

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN} \tag{3.8}$$

and then the sum equation

$$\begin{aligned} \sum_{j=0}^{N-1}\left\|\hat{u}[j+k|k]\right\|_R^2 = \hat{u}^T[k|k]R\hat{u}[k|k] + \hat{u}^t[k+1|k]R\hat{u}[k+1|k] + \cdots \\ + \hat{u}^T[k+N-1|k]R\hat{u}[k+N-1|k] \end{aligned} \tag{3.9}$$

becomes

$$\underbrace{\begin{bmatrix} \hat{u}^T[k|k] & \hat{u}^T[k+1|k] & \cdots & \hat{u}^T[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}^T} \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \underbrace{\begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}}_{\hat{\mathbf{u}}} \tag{3.10}$$

Using the notation

$$\mathbf{R} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} = diag_N(R) \tag{3.11}$$

we get to

$$\sum_{j=0}^{N-1} \left\| \hat{u}[j] \right\|_R^2 = \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \tag{3.12}$$

Similarly, we can write

$$\begin{gathered} \left\| \hat{x}[k+N|k] \right\|_{P_f}^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = x^T[k|k]Qx[k|k] \\ + \hat{x}^T[k+1|k]Q\hat{x}[k+1|k] + \cdots + \hat{x}^T[k+N-1|k]Q\hat{x}[k+N-1|k] + \hat{x}^T[k+N|k]P_f\hat{x}[k+N|k] \\ = \hat{x}^T[k|k]Q\hat{x}[k|k] + \\ \underbrace{\begin{bmatrix} \hat{x}^T[k+1|k] & \hat{x}^T[k+2|k] & \cdots & \hat{x}^T[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]^T} \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} \underbrace{\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix}}_{\hat{\mathbf{x}}[k]} \end{gathered} \tag{3.13}$$

with the notation

$$Q = \begin{bmatrix} Q & 0 & \cdots & 0 \\ 0 & Q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & P_f \end{bmatrix} = \begin{bmatrix} diag_{N-1}(Q) & 0 \\ 0 & P_f \end{bmatrix} \in \mathbb{R}^{nN \times nN} \tag{3.14}$$

as

$$\left\| \hat{x}[k+N|k] \right\|_P^2 + \sum_{j=0}^{N-1} \left\| \hat{x}[k+j|k] \right\|_Q^2 = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 \tag{3.15}$$

So, the cost function can be written in the form

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \left\| x[k] \right\|_P^2 + \left\| \hat{\mathbf{x}} \right\|_{\mathbf{Q}}^2 + \left\| \hat{\mathbf{u}} \right\|_{\mathbf{R}}^2 \tag{3.16}$$

3.2 Restrictions

The restrictions are introduced aiming to drive the states of the system to the origin while solving

$$\min_{\hat{\mathbf{x}}\in\mathbb{R}^{nN},\hat{\mathbf{u}}\in\mathbb{R}^{pN}} \mathbf{J}(\hat{\mathbf{x}},\hat{\mathbf{u}}) \tag{3.17}$$

respecting constraints applied on

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathcal{U}\times\mathcal{U}\times\cdots\times\mathcal{U} = \mathcal{U}^N \tag{3.18}$$

$$\hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathcal{X}\times\mathcal{X}\times\cdots\times\mathcal{X}_f = \mathcal{X}^{N-1}\times\mathcal{X}_f \tag{3.19}$$

$\mathcal{X}_f$ is the maximum invariant and admissible set (MAS) with respect to the terminal state and control constraints under the action of a terminal control force.

$$\{\hat{x}\in\mathbb{R}^n : S_f\hat{x}\leq b_f\} \tag{3.20}$$

The constraint on $\hat{\mathbf{u}}$ can be written as

$$\begin{bmatrix} S_u & 0 & \cdots & 0 \\ 0 & S_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_u \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \leq \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \end{bmatrix} \tag{3.21}$$

or, in a shorter form

$$\mathbf{S_u}\hat{\mathbf{u}} \leq \mathbf{b_u} \tag{3.22}$$

with

$$\mathbf{S_u} = diag_N(S_u), \mathbf{b_u} = col_N(b_u) \tag{3.23}$$

Similarly, the constraints on $\hat{\mathbf{x}}$ can be written as

$$\begin{bmatrix} S_x & 0 & \cdots & 0 \\ 0 & S_x & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_x \end{bmatrix} \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \leq \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \end{bmatrix} \tag{3.24}$$

or, in short

$$\mathbf{S_x}\hat{\mathbf{x}} \leq \mathbf{b_x} \tag{3.25}$$

with

$$\mathbf{S_x} = \begin{bmatrix} diag_{N-1}(S_x) & 0 \\ 0 & S_f \end{bmatrix}, \mathbf{b_x} = \begin{bmatrix} col_{N-1}(b_x) \\ b_f \end{bmatrix} \tag{3.26}$$

3.3 Prediction Equation

Let's express the relationship between the vectors

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \tag{3.27}$$

based on the state equation

$$\hat{x}[k+i+1|k] = A\hat{x}[k|k] + B\hat{u}[k|k] \tag{3.28}$$

and initial condition $x[k]$.

The solution for the state equation is:

$$\hat{x}[k+i|k] = A^i\hat{x}[k|k] + \sum_{j=0}^{i-1} A^{i-1-j}B\hat{u}[k+j|k] \tag{3.29}$$

and it can be written for $k = 1, 2, ..., N$ as

$$\hat{x}[k+1|k] = Ax[k] + B\hat{u}[k|k] \tag{3.30}$$

$$\hat{x}[k+2|k] = A^2x[k] + AB\hat{u}[k|k] + B\hat{u}[k+1|k] \tag{3.31}$$

$$\vdots \tag{3.32}$$

$$\hat{x}[k+N|k] = A^Nx[k] + A^{N-1}B\hat{u}[k|k] + \cdots + AB\hat{u}[k+N-2|k] + B\hat{u}[k+N-1|k] \tag{3.33}$$

or

$$\begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \tag{3.34}$$

This can be written in a shorter form as

$$\hat{\mathbf{x}} = \mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}} \tag{3.35}$$

with

$$\mathbf{A} = \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^N \end{bmatrix}, \mathbf{B} = \begin{bmatrix} B & 0 & \cdots & 0 \\ AB & B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \tag{3.36}$$

3.4 Optimization Problem

We can rewrite the cost function as

$$J = \left\|x[k|k]\right\|_P^2 + \overbrace{\left\|\hat{\mathbf{x}}\right\|_{\mathbf{Q}}^2}^{\square} + \left\|\hat{\mathbf{u}}\right\|_{\mathbf{R}}^2 \tag{3.37}$$

$$\begin{aligned} \square &= \left\|\hat{\mathbf{x}}\right\|_{\mathbf{Q}}^2 && (3.38) \\ &= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}\right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}\right] && (3.39) \\ &= \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}\right]^T \mathbf{Q} \left[\mathbf{A}x[k|k] + \mathbf{B}\hat{\mathbf{u}}\right] && (3.40) \\ &= \left\|\mathbf{A}x[k|k]\right\|_{\mathbf{Q}}^2 + \left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}} \\ &\quad + \hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k] + \hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{B}\hat{\mathbf{u}} && (3.41) \end{aligned}$$

replacing $\square$ back in the cost equation and reordering the elements

$$\begin{aligned} J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = {} & \left\|x[k|k]\right\|_P^2 + \left\|\mathbf{A}x[k|k]\right\|_{\mathbf{Q}}^2 && (3.42) \\ & + \left[\mathbf{A}x[k|k]\right]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}} + \hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k] && (3.43) \\ & + \hat{\mathbf{u}}^T\mathbf{R}\hat{\mathbf{u}} + \hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{B}\hat{\mathbf{u}} && (3.44) \end{aligned}$$

The constant part will be denoted by c_0

Recalling that the matrix $\mathbf{Q}$ is symmetric, we have:

$$\Big[\mathbf{A}x[k|k]\Big]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}} = \Big\{\hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x[k|k]\Big\}^T \tag{3.45}$$

As these terms are scalars, it can be concluded that they are equal to each other, therefore:

$$\begin{aligned}&\Big[\mathbf{A}x[k|k]\Big]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}} + \hat{\mathbf{u}}^T\mathbf{B}^T\mathbf{Q}\mathbf{A}x(0)\\ &= 2\Big[\mathbf{A}x[k|k]\Big]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}}\end{aligned} \tag{3.46}$$

Finally, it can be written

$$J(\hat{\mathbf{u}}) = \hat{\mathbf{u}}^T\Big(\mathbf{R} + \mathbf{B}^T\mathbf{Q}\mathbf{B}\Big)\hat{\mathbf{u}} + 2\Big[\mathbf{A}x[k|k]\Big]^T \mathbf{Q}\mathbf{B}\hat{\mathbf{u}} + c_0 \tag{3.47}$$

with

$$c_0 = \Big\|x(0)\Big\|_P^2 + \Big\|\mathbf{A}x(0)\Big\|_{\mathbf{Q}}^2 \tag{3.48}$$

$$c_0 = \Big\|x[k|k]\Big\|_P^2 + \Big\|\mathbf{A}x[k|k]\Big\|_{\mathbf{Q}}^2$$

Note that we will now use the notation $J(\hat{\mathbf{u}})$ instead of $J(\hat{\mathbf{x}}, \hat{\mathbf{u}})$, since $\hat{\mathbf{x}}$ has been eliminated from the cost expression.

To complete the reformulation of the optimization problem in terms of $\hat{\mathbf{u}}$, the relation between $\hat{\mathbf{x}}$ and $\hat{\mathbf{u}}$ in the **restrictions** must be considered.

The optimization problem can be expressed as

$$\min_{\hat{\mathbf{x}}\in\mathbb{R}^{nN}, \hat{\mathbf{u}}\in\mathbb{R}^{pN}} J(\hat{x}, \hat{u}) \tag{3.49}$$

where

$$J(\hat{\mathbf{x}}, \hat{\mathbf{u}}) = \Big\|x[k|k]\Big\|_P^2 + \Big\|\hat{\mathbf{x}}\Big\|_{\mathbf{Q}}^2 + \Big\|\hat{\mathbf{u}}\Big\|_{\mathbf{R}}^2 \tag{3.50}$$

with

$$\hat{\mathbf{u}}[k] = \begin{bmatrix} \hat{u}[k|k] \\ \hat{u}[k+1|k] \\ \vdots \\ \hat{u}[k+N-1|k] \end{bmatrix} \in \mathbb{R}^{pN}, \hat{\mathbf{x}}[k] = \begin{bmatrix} \hat{x}[k+1|k] \\ \hat{x}[k+2|k] \\ \vdots \\ \hat{x}[k+N|k] \end{bmatrix} \in \mathbb{R}^{nN}, \tag{3.51}$$

subject to

$$\mathbf{S}_{\mathbf{x}}\hat{\mathbf{x}}[k] \leq \mathbf{b}_{\mathbf{x}} \tag{3.52}$$

$$\mathbf{S_u}\hat{\mathbf{u}}[k] \leq \mathbf{b_u} \tag{3.53}$$

The vectors $\hat{\mathbf{x}}[k|k]$ and $\hat{\mathbf{u}}[k|k]$ are linked by the following **prediction equation**:

$$\hat{\mathbf{x}}[k+1|k] = \mathbf{A}\hat{x}[k|k] + \mathbf{B}\hat{\mathbf{u}}[k|k] \tag{3.54}$$

Thus, the cost can be rewritten as

$$\begin{gathered} J(\hat{\mathbf{u}}[k]) = \hat{\mathbf{u}}[k]^T(\mathbf{R} + \mathbf{B}^T\mathbf{Q}\mathbf{B})\hat{\mathbf{u}}[k] \\ 2\Big[\mathbf{A}\hat{x}[k|k]\Big]^T\mathbf{Q}\mathbf{B}\hat{\mathbf{u}}[k] + c_0 \end{gathered}$$

with

$$c_0 = \Big\|\hat{x}[k|k]\Big\|_P^2 + \Big\|\mathbf{A}\hat{x}[k|k]\Big\|_{\mathbf{Q}}^2$$

and the constraints can be expressed as

$$\mathbf{S}\hat{\mathbf{u}}[k] \leq \mathbf{b} \tag{3.55}$$

with

$$\mathbf{S} = \begin{bmatrix} \mathbf{S_x}\mathbf{B} \\ \mathbf{S_u} \end{bmatrix}, \mathbf{b} = \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x}\mathbf{A}\hat{x}[k|k] \\ \mathbf{b_u} \end{bmatrix} \tag{3.56}$$

Having obtained the solution $\hat{\mathbf{u}}[k|k]$ for the problem $\mathbb{P}\Big(\hat{x}[k|k]\Big)$, the control to be applied to the plant is given by

$$\hat{u}[k|k] = \hat{u}^*[k|k]$$

where $\hat{u}^*[k|k]$ corresponds to the first p elements of $\hat{\mathbf{u}}^*[k|k]$.

3.5 Feasibility Region

slide: aula 8 slide 30

Pergunta: Para m dado N, como determinar o conjunto $\mathcal{D} \subset \mathcal{X}$ de condições iniciais $x(0)$ a partir das quais o conjunto alvo $\mathcal{X}_f$ pode ser atingido repeitando as restricoes sobre o estado e o controle ?

Alternativamente: Como determinal o conjunto $\mathcal{D}$ de condicoes iniciais $x(0)$ para as quais o problema de otimizacao e factivel

$\mathcal{D}$ é o conjunto (ou regiao) de pontos $x(0) \in \mathcal{X}$ para os quais exist $\mathbf{u} \in \mathbb{R}^{pN}$ que satisfaz as sequintes restricoes:

$$\begin{bmatrix} \mathbf{S_x B} \\ \mathbf{S_u} \end{bmatrix} \mathbf{u} \leq \begin{bmatrix} \mathbf{b_x} - \mathbf{S_x A} x(0) \\ \mathbf{b_u} \end{bmatrix} \tag{3.57}$$

Essas restricoes podem ser reescritas como

$$\begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \end{bmatrix} \tag{3.58}$$

ou ainda, acrescentando-se a restricao $S_x x(0) \leq b_x$,

$$\begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \\ 0 & S_x \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix} \leq \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \\ b_x \end{bmatrix} \tag{3.59}$$

De forma resumida, pode-se escrever

$$S_w w \leq b_w \tag{3.60}$$

com

$$w = \begin{bmatrix} \mathbf{u} \\ x(0) \end{bmatrix}, S_w = \begin{bmatrix} \mathbf{S_x B} & \mathbf{S_x A} \\ \mathbf{S_u} & 0 \\ 0 & S_x \end{bmatrix}, b_w = \begin{bmatrix} \mathbf{b_x} \\ \mathbf{b_u} \\ b_x \end{bmatrix} \tag{3.61}$$

A restricao $S_w w \leq b_w$ define um poliedro $\mathcal{P}_w$ no espaco $\mathbb{R}^{(pN+n)}$ correspondente as variaveis $\mathbf{u}$, x(0).

O conjunto $\mathcal{D}$ e obtido **projetando-se** $\mathcal{P}_w$ sobre o subespaco de suas ultimas n componentes.

A **projecao** pose ser realizada empregando a funcao projection do MPT Toolbox:

$$\boxed{Sw = [Sxn * Bn]...}$$

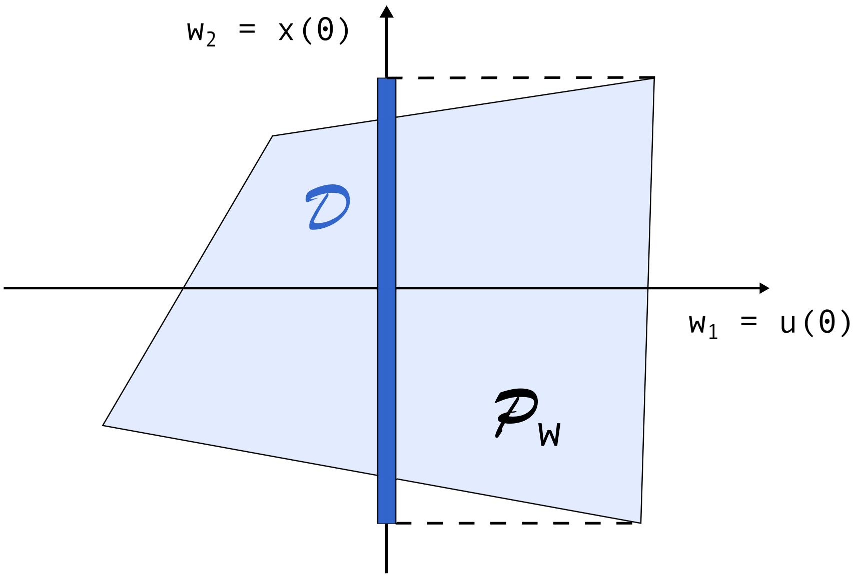

FIGURE 3.1 – Ilustracao para n = 1 estado, p = 1 entrada e N = 1 passo

4 Predictive Control for Stabilization of Periodic Trajectories

4.1 Statement of the problem

Considering a plant of an affine switched system described by:

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \tag{4.1}$$

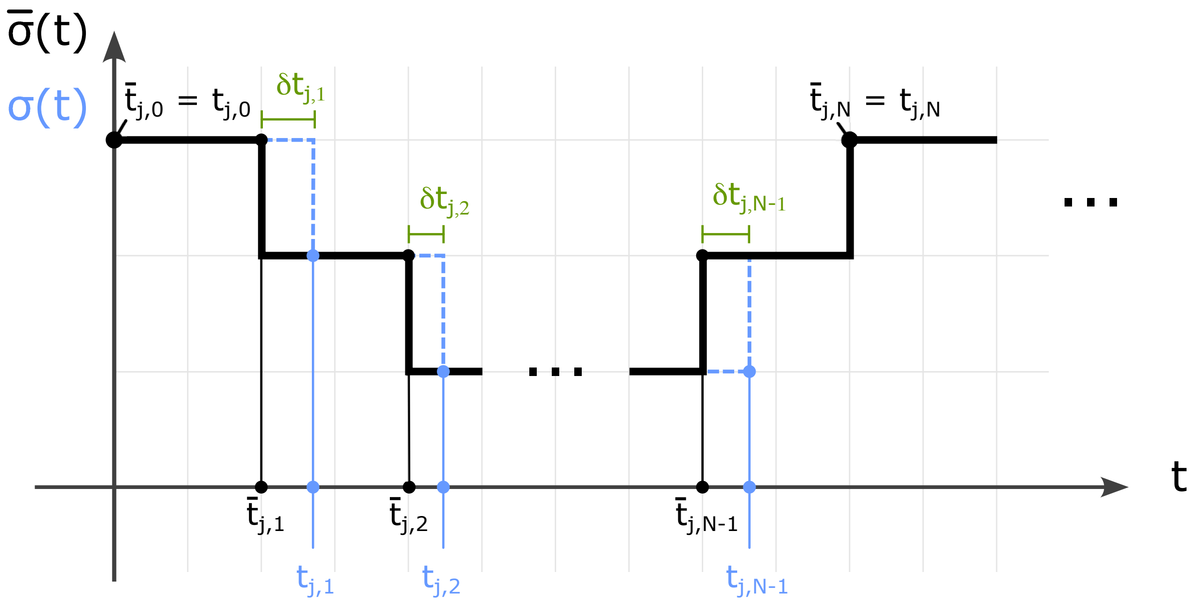

FIGURE 4.1 – Switching signals

We assume the existence of a periodic switching signal $\bar{\sigma}(t)$ associated with a periodic trajectory $\bar{x}(t)$, as obtained in Chapter 2. Our objective is to introduce perturbations in the switching instants such that the trajectory $x(t)$ converges to the reference trajectory $\bar{x}(t)$.

Figure 4.1 displays the jth cycle of the switching signals $\bar{\sigma}(t)$ and $\sigma(t)$, which comprise N switching actions. Within this cycle, the ith switching times of $\bar{\sigma}(t)$ and $\sigma(t)$ are denoted by $\bar{t}_{j,i}$ and $t_{j,i}$, respectively. The relation between $\bar{t}_{j,i}$ and $t_{j,i}$ is given by

$$t_{j,i} = \bar{t}_{j,i} + \delta t_{j,i} \tag{4.2}$$

To do this using predictive control, we need to find a linear equation that relates the switching times and the resulting system at the end of one cycle.

4.2 Linearization procedure

The solution for the state equation (4.1) at time t_1, starting from the initial condition $x(t_0)$, is of the form

$$x(t_1) = e^{A_1(t_1 - t_0)}x(t_0) + \int_{t_0}^{t_1} e^{A_1(t_1-\tau)}B_1 d\tau \tag{4.3}$$

$$\bar{x}(\bar{t}_1) = e^{A_1(\bar{t}_1 - \bar{t}_0)}\bar{x}(\bar{t}_0) + \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau \tag{4.4}$$

subtracting (4.4) from (4.3) we get

$$\begin{aligned} x(t_1) - \bar{x}(\bar{t}_1) = {} & \overbrace{e^{A_1(t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0)}^{\square} \\ & \underbrace{+\int_{t_0}^{t_1} e^{A_1(t_1-\tau)}B_1 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)}B_1 d\tau}_{\bigcirc} \end{aligned} \tag{4.5}$$

expanding (4.5) first in $\square$:

$$\begin{aligned} \square &= e^{A_1(t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1+\delta t_1-t_0)}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= e^{A_1(\bar{t}_1-t_0)}e^{\delta t_1}x(t_0) - e^{A_1(\bar{t}_1-\bar{t}_0)}\bar{x}(\bar{t}_0) \\ &= \bar{F}_1 e^{\delta t_1}x(t_0) - \bar{F}_1\bar{x}(\bar{t}_0) \\ &= \bar{F}_1(I + A_1\delta t_1)x(t_0) - \bar{F}_1\bar{x}(\bar{t}_0) \\ &= \bar{F}_1[x(t_0) - \bar{x}(t_0)] + \bar{F}_1 A_1 \delta t_1 \end{aligned} \tag{4.6}$$

and then in $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_0}^{t_1} e^{A_1(t_1-\tau)} B_1 d\tau - \int_{\bar{t}_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
&= \int_{t_0}^{\bar{t}_1+\delta t_1} e^{A_1(\bar{t}_1+\delta t_1-\tau)} B_1 d\tau - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
&= e^{A_1\delta t_1}\left[\int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau + \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_1(t_{r_1}-\tau)} B_1 d\tau\right] - \int_{t_0}^{\bar{t}_1} e^{A_1(\bar{t}_1-\tau)} B_1 d\tau \\
&= e^{A_1\delta t_1}[\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
&= (I + A_1\delta t_1)[\bar{G}_1 + B_1\delta t_1] - \bar{G}_1 \\
&= \cancel{\bar{G}_1} + B_0\delta t_1 + A_1\bar{G}_1\delta t_1 + \cancel{A_1B_1\delta t_1\delta t_1} - \cancel{\bar{G}_1} \\
&= A_1\bar{G}_1\delta t_1 + B_1\delta t_1
\end{aligned} \tag{4.7}
$$

the resulting equations (4.6) and (4.7) are replaced back in (4.5) to complete the difference equation

$$
\boxed{x(t_1) - \bar{x}(\bar{t}_1) = \bar{F}_1[x(t_0) - \bar{x}(t_0)] + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1} \tag{4.8}
$$

Repeating the steps for $x(t_2)$ we have

$$
x(t_2) = e^{A_2(t_2-t_1)}x(t_1) + \int_{t_1}^{t_2} e^{A_2(t_2-\tau)} B_2 d\tau \tag{4.9}
$$

$$
\bar{x}(\bar{t}_2) = e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) + \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau \tag{4.10}
$$

subtracting (4.10) from (4.9)

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) &= \overbrace{e^{A_2(t_2-t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1)}^{\square} \\
&\quad \underbrace{+ \int_{t_1}^{t_2} e^{A_2(t_2-\tau)} B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)} B_2 d\tau}_{\bigcirc}
\end{aligned} \tag{4.11}
$$

expanding $\square$ in (4.11):

$$
\begin{aligned}
\square &= e^{A_2(\bar{t}_2+\delta t_2-\bar{t}_1-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= e^{A_2(\bar{t}_2-\bar{t}_1)}e^{A_2(\delta t_2-\delta t_1)}x(t_1) - e^{A_2(\bar{t}_2-\bar{t}_1)}\bar{x}(\bar{t}_1) \\
&= \bar{F}_2[I + A_2(\delta t_2 - \delta t_1)]x(t_1) - \bar{F}_2\bar{x}(\bar{t}_1) \\
&= \bar{F}_2[x(t_1) - \bar{x}(\bar{t}_1)] + \bar{F}_2 A_2 x(t_1)\delta t_2 - \bar{F}_2 A_2 x(t_1)\delta t_1
\end{aligned} \tag{4.12}
$$

and expanding $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_1}^{t_2} e^{A_2(t_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= \int_{\bar{t}_1+\delta t_1}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2+\delta t_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= e^{A_2\delta t_2}\left[\int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau\right] \\
&\quad - \int_{\bar{t}_1}^{\bar{t}_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - \int_{\bar{t}_1}^{\bar{t}_1+\delta t_1} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau + \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_2(\bar{t}_2-\tau)}B_2 d\tau\right] - \bar{G}_2 \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - e^{A_2(\bar{t}_2-\bar{t}_1)}B_2\delta t_1 + B_2\delta t_2\right] - \bar{G}_2 \\
&= (I + A_2\delta t_2)\left[\bar{G}_2 - \bar{F}_2 B_2\delta t_1 + B_2\delta t_2\right] - \bar{G}_2 \\
&= \cancel{\bar{G}_2} - \bar{F}_2 B_2\delta t_1 + B_2\delta t_2 + A_2\bar{G}_2\delta t_2 - \cancel{\bar{G}_2} \\
&= A_2\bar{G}_2\delta t_2 + B_2\delta t_2 - \bar{F}_2 B_2\delta t_1
\end{aligned} \tag{4.13}
$$

and replacing (4.12) and (4.13) back into equation (4.11) we get to the difference equation for $x(t_2) - \bar{x}(\bar{t}_2)$

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2[x(t_1) - \bar{x}_1(\bar{t}_1)] \\
& - \bar{F}_2[A_2 x(t_1) + B_2]\delta t_1 \\
& + [\bar{F}_2 A_2 \underbrace{x(t_1)}_{\triangle} + A_2\bar{G}_2 + B_2]\delta t_2
\end{aligned} \tag{4.14}
$$

from (4.14), $\triangle$ can be expanded by replacing the resulting equation of (4.8)

$$
\triangle = x(t_1) = \bar{x}(\bar{t}_1) + \bar{F}_1[x(t_0) - \bar{x}(t_0)] + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1 \tag{4.15}
$$

note that when $\triangle$ is replaced in eq. (4.14) it will produce a product with δt_1 and δt_2 like

$$
x(t_1)\delta t_1 = \bar{x}(\bar{t}_1)\delta t_1 + \bar{F}_1\overbrace{[x(t_0) - \bar{x}(t_0)]\delta t_1}^{\square} + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\overbrace{\delta t_1\delta t_1}^{\bigcirc}
$$

both $\square$ and $\bigcirc$ are products of differences that produces a smaller number and they will be neglected in the first order aproximations

$$
x(t_1)\delta t_1 \doteq \bar{x}(\bar{t}_1)\delta t_1 \tag{4.16}
$$

replacing (4.8) and (4.15) in (4.14) it follows that

$$
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2\{\bar{F}_1[x(t_0) - \bar{x}(t_0) + [\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 + B_1]\delta t_1]\} \\
&- \bar{F}_2[A_2\bar{x}(t_1) + B_2]\delta t_1 \\
&+ [\bar{F}_2 A_2 \bar{x}(t_1) + A_2\bar{G}_2 + B_2]\delta t_2
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
x(t_2) - \bar{x}(\bar{t}_2) =& \bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\
&+ \bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\
&+ [\bar{F}_2 A_2 \bar{x}(\bar{t}_1) + A_2\bar{G}_2 + B_2]\delta t_2
\end{aligned}
}
\tag{4.17}
$$

Repeating the procedure for $x(t_3)$ we have

$$
x(t_3) = e^{A_3(t_3 - t_2)}x(t_2) + \int_{t_2}^{t_3} e^{A_3(t_3-\tau)}B_3 d\tau \tag{4.18}
$$

$$
\bar{x}(\bar{t}_3) = e^{A_3(\bar{t}_3 - \bar{t}_2)}\bar{x}(\bar{t}_2) + \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau \tag{4.19}
$$

subtracting (4.18) from (4.19)

$$
\begin{aligned}
x(t_3) - > -\bar{x}(\bar{t}_3) = & \overbrace{e^{A_3(t_3-t_2)}x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2)}^{\square} \\
& \underbrace{+ \int_{t_2}^{t_3} e^{A_3(t_3-\tau)}B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)}B_3 d\tau}_{\bigcirc}
\end{aligned}
\tag{4.20}
$$

expanding $\square$ in (4.20):

$$
\begin{aligned}
\square &= e^{A_3(\bar{t}_3 + \delta t_3 - \bar{t}_2 - \delta t_2)}x(t_2) - e^{A_3(\bar{t}_3 - \bar{t}_2)}\bar{x}(\bar{t}_2) \\
&= e^{A_3(\bar{t}_3-\bar{t}_2)}e^{A_3(\delta t_3 - \delta t_2)}x(t_2) - e^{A_3(\bar{t}_3-\bar{t}_2)}\bar{x}(\bar{t}_2) \\
&= \bar{F}_3[I + A_3(\delta t_3 - \delta t_2)]x(t_2) - \bar{F}_3\bar{x}(\bar{t}_2) \\
&= \bar{F}_3[x(t_2) - \bar{x}(\bar{t}_2)] + \bar{F}_3 A_3 x(t_2)\delta t_3 - \bar{F}_3 A_3 x(t_2)\delta t_2
\end{aligned}
\tag{4.21}
$$

and expanding $\bigcirc$:

$$
\begin{aligned}
\bigcirc &= \int_{t_2}^{t_3} e^{A_3(t_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
&= \int_{\bar{t}_2+\delta t_2}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3+\delta t_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
&= e^{A_3\delta t_3}\left[\int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau\right] \\
&\quad - \int_{\bar{t}_2}^{\bar{t}_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - \int_{\bar{t}_2}^{\bar{t}_2+\delta t_2} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau + \int_{\bar{t}_3}^{\bar{t}_3+\delta t_3} e^{A_3(\bar{t}_3-\tau)} B_3 d\tau\right] - \bar{G}_3 \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - e^{A_3(\bar{t}_3-\bar{t}_2)} B_3\delta t_2 + B_3\delta t_3\right] - \bar{G}_3 \\
&= (I + A_3\delta t_3)\left[\bar{G}_3 - \bar{F}_3 B_3\delta t_2 + B_3\delta t_3\right] - \bar{G}_3 \\
&= \cancel{\bar{G}_3} - \bar{F}_3 B_3\delta t_2 + B_3\delta t_3 + A_3\bar{G}_3\delta t_3 - \cancel{\bar{G}_3} \\
&= A_3\bar{G}_3\delta t_3 + B_3\delta t_3 - \bar{F}_3 B_3\delta t_2
\end{aligned} \tag{4.22}
$$

and then replacing (4.21) in (4.22) back in eq. (4.20) we get to the difference equation for $x(t_3) - \bar{x}(\bar{t}_3)$

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3[x(t_2) - \bar{x}_2(\bar{t}_2)] \\
& - \bar{F}_3[A_3 x(t_2) + B_3]\delta t_2 \\
& + [\bar{F}_3 A_3 x(t_2) + A_3\bar{G}_3 + B_3]\delta t_3
\end{aligned} \tag{4.23}
$$

from (4.17), $\triangle$ can be expanded the same way as in the past iteration and $x(t_2) - \bar{x}(\bar{t}_2)$ can be replaced in (4.23)

$$
\begin{aligned}
x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3\{\bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\
& + [\bar{F}_2 A_2\bar{x}(t_1) + A_2\bar{G}_2 + B_2]\delta t_2\} \\
& - \bar{F}_3[A_3\bar{x}(t_2) + B_3]\delta t_2 \\
& + [\bar{F}_3 A_3\bar{x}(t_2) + A_3\bar{G}_3 + B_3]\delta t_3
\end{aligned}
$$

after rearranging the terms, one arrives at

$$\boxed{\begin{aligned} x(t_3) - \bar{x}(\bar{t}_3) =& \bar{F}_3\bar{F}_2\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_3\bar{F}_2[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\ &+ \bar{F}_3[\bar{F}_2 A_2 \bar{x}(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &+ [\bar{F}_3 A_3 \bar{x}(t_2) + A_3\bar{G}_3 + B_3]\delta t_3 \end{aligned}} \tag{4.24}$$

Now, there are enough iterations to deduce a pattern on each increment of the equation, so the equation for $x(t_i)$ can be deduced as

$$\boxed{\begin{aligned} x(t_i) - \bar{x}(\bar{t}_i) =& \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_1[\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2]\delta t_1 \\ &+ \bar{F}_i\bar{F}_{i-1}\ldots\bar{F}_2[\bar{F}_2 A_2 x(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &\quad\vdots \\ &+ \bar{F}_i[\bar{F}_{i-1} A_{i-1} \bar{x}(t_{i-2}) + A_{i-1}\bar{G}_{i-1} - A_i\bar{x}(\bar{t}_{i-1}) + B_{i-1} - B_i]\delta t_{i-1} \\ &+ [\bar{F}_i A_i \bar{x}(t_i) + A_i\bar{G}_i + B_i]\delta t_i \end{aligned}} \tag{4.25}$$

and then, for $x(t_N)$ the equation is as follows

$$\begin{aligned} x(t_N) - \bar{x}(\bar{t}_N) =& \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_1[x(t_0) - \bar{x}(t_0)] \\ &+ \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_2[\overbrace{\bar{F}_1 A_1 x(t_0) + A_1\bar{G}_1 - A_2\bar{x}(\bar{t}_1) + B_1 - B_2}^{\square}]\delta t_1 \\ &+ \bar{F}_N\bar{F}_{N-1}\ldots\bar{F}_3[\bar{F}_2 A_2 x(t_1) + A_2\bar{G}_2 - A_3\bar{x}(\bar{t}_2) + B_2 - B_3]\delta t_2 \\ &\quad\vdots \\ &+ \bar{F}_N[\bar{F}_{N-1} A_{N-1} \bar{x}(t_{N-2}) + A_{N-1}\bar{G}_{N-1} - A_N\bar{x}(\bar{t}_{N-1}) \\ &\qquad + B_{N-1} - B_N]\delta t_{N-1} \\ &+ [\underbrace{\bar{F}_N A_{N-1} \bar{x}(t_{N-1}) + A_N\bar{G}_N + B_N}_{\bigcirc}]\delta t_N \end{aligned} \tag{4.26}$$

The equation (4.26) can be developed further to remove the dependencies on the intermediary states $x(t_0), x(t_1), ..., x(t_{N-2})$. By expanding $\square$ and $\bigcirc$, the following identity holds

$$
\begin{aligned}
\square &= \bar{F}_1 A_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 \bar{F}_1 x(t_0) + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 [\bar{x}(\bar{t}_1) - \bar{G}_1] + A_1 \bar{G}_1 - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= A_1 \bar{x}(\bar{t}_1) - \cancel{A_1 \bar{G}_1} + \cancel{A_1 \bar{G}_1} - A_2 \bar{x}(\bar{t}_1) + B_1 - B_2 \\
&= (A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2
\end{aligned} \tag{4.27}
$$

$$
\begin{aligned}
\bigcirc &= \bar{F}_N A_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
&= A_N \bar{F}_N \bar{x}(t_{N-1}) + A_N \bar{G}_N + B_N \\
&= A_N [\bar{x}(\bar{t}_N) - \bar{G}_N] + A_N \bar{G}_N + B_N \\
&= A_N \bar{x}(\bar{t}_N) - \cancel{A_N \bar{G}_N} + \cancel{A_N \bar{G}_N} + B_N \\
&= A_N \bar{x}(\bar{t}_N) + B_N
\end{aligned} \tag{4.28}
$$

replacing (4.27) and (4.28) in (4.26)

$$
\boxed{
\begin{aligned}
x(t_N) - \bar{x}(\bar{t}_N) = \quad & \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 [x(t_0) - \bar{x}(t_0)] \\
& + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2 [(A_1 - A_2) \bar{x}(\bar{t}_1) + B_1 - B_2] \delta t_1 \\
& + \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3 [(A_2 - A_3) \bar{x}(\bar{t}_2) + B_2 - B_3] \delta t_2 \\
& \qquad \vdots \\
& + \bar{F}_N [(A_{N-1} - A_N) \bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N] \delta t_{N-1} \\
& + [A_N \bar{x}(\bar{t}_N) + B_N] \delta t_N
\end{aligned}
} \tag{4.29}
$$

and renaming the terms of (4.29) to

$$e(t) = x(t) - \bar{x}(\bar{t}) \tag{4.30}$$

$$\Phi = \bar{F}_N \bar{F}_{N-1} \dots \bar{F}_1 \tag{4.31}$$

$$\begin{aligned}\Gamma = \Big[&\bar{F}_N \bar{F}_{N-1} \dots \bar{F}_2[(A_1 - A_2)\bar{x}(\bar{t}_1) + B_1 - B_2],\\ &\bar{F}_N \bar{F}_{N-1} \dots \bar{F}_3[(A_2 - A_3)\bar{x}(\bar{t}_2) + B_2 - B_3],\\ &\quad\vdots\\ &\bar{F}_N[(A_{N-1} - A_N)\bar{x}(\bar{t}_{N-1}) + B_{N-1} - B_N],\\ &A_N \bar{x}(\bar{t}_N) + B_N\Big]\end{aligned} \tag{4.32}$$

$$\delta t = [\delta t_1, \delta t_2, \dots, \delta t_N]^T \tag{4.33}$$

we get to the affine equation

$$\boxed{e(t_N) = \Phi e(t_0) + \Gamma \delta t} \tag{4.34}$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \tag{4.35}$$

where $\bar{x}$ is calculated as

$$\bar{x}(\bar{t}_1) = F_1 \bar{x}(\bar{t}_0) + G_1 \tag{4.36}$$

$$\bar{x}(\bar{t}_2) = F_2 \bar{x}(\bar{t}_1) + G_2 \tag{4.37}$$

$$\vdots$$

$$\bar{x}(\bar{t}_N) = F_N \bar{x}(\bar{t}_{N-1}) + G_N \tag{4.38}$$

and F_i and G_i are calculated as

$$[F_i, G_i] = c2dm(Ai, Bi, [], [], dti, \text{`}zoh\text{`}) \tag{4.39}$$

4.3 MPC Problem

Note that the equation (4.34) relates to the equation (3.1) as both describe the evolution of a system over time. The equation (3.1) represents the system's state and input at a given time step, and the equation (4.34) describes the error between the system's predicted and actual values at two different time points.

Referring to equation (3.1), we can make some substitutions to simplify our discussion. We'll replace variable x with e, and u with δt. Additionally, we'll denote the cycle index with k. With these substitutions in place, we arrive at equation (4.34).

4.3.1 Cost Function

$$J = \left\| e[N_p] \right\|_{P_f}^2 + \sum_{j=0}^{N_p-1} \left\| e[j] \right\|_Q^2 + \left\| \delta t[j] \right\|_R^2 \tag{4.40}$$

$$e(t_{j,N}) = \Phi e(t_{j,0}) + \Gamma \delta t[j] \tag{4.41}$$

where $Q = Q^T > 0$, $\rho > 0$

$$\delta t[j] = \left[t_{j,1} - \bar{t}_{j,1}, t_{j,2} - \bar{t}_{j,2}, ..., t_{j,N-1} - \bar{t}_{j,N-1} \right]^T \tag{4.42}$$

4.3.2 Dwell-time Constraints

We want to enforce a constraint on the dwell times where the switching times will the subject to the constraints:

$$(t_i - t_{i-1}) \geq \Delta t_{i,min} \ , \quad i = 1, 2, \dots, N \tag{4.43}$$

then we can write the restriction as

$$\delta t_1 \geq -\Delta \bar{t}_1 + \Delta t_{1,min} \tag{4.44}$$

$$\delta t_i - \delta t_{i-1} \geq -\Delta \bar{t}_i + \Delta t_{i,min} \ , \quad i = 1, 2, \dots, N \tag{4.45}$$

$$\delta t_{N-1} \geq -\Delta \bar{t}_N + \Delta t_{N,min} \tag{4.46}$$

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1} \tag{4.47}$$

to match the constraint from (3.55)

$$\mathbf{S}\hat{\mathbf{u}}[k] \leq b$$

the equations can be written as

$$-\delta t_1 \leq +\Delta\bar{t}_1 - \Delta t_{1,min} \tag{4.48}$$

$$-\delta t_i + \delta t_{i-1} \leq +\Delta\bar{t}_i - \Delta t_{i,min}\ , \quad i = 1, 2, \ldots, N \tag{4.49}$$

$$-\delta t_{N-1} \leq +\Delta\bar{t}_N - \Delta t_{N,min} \tag{4.50}$$

with

$$\mathbf{L} = \begin{bmatrix} -1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & -1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \tag{4.51}$$

$$c = \begin{bmatrix} \Delta t_{r_1} - \Delta t_{r_{1,min}} \\ \Delta t_{r_2} - \Delta t_{r_{2,min}} \\ \vdots \\ \Delta t_{r_{N-1}} - \Delta t_{r_{N-1,min}} \\ \Delta t_{r_N} - \Delta t_{r_{N,min}} \end{bmatrix} \tag{4.52}$$

4.4 Application Examples

4.4.1 Double Integrator

4.4.1.1 Parameters of the Simulation

- operation modes of the system (Ω): $\left[1, 2, 3, 4\right]$,
- maximum number of operation modes (s_{max}): 4,
- dynamics of the system:

$$A_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_3 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad A_4 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \qquad b_4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- time step of the simulation (t_{step}): set to 0.01,
- reference state vector (x_{ref}): $\left[2, \text{-}1\right]$,
- weighting matrix for the trajectory computation (Q):

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- maximum value for the period of the trajectory cycle ($T_{p_{max}}$): 1,
- time sequence (T_s): $\left[0, 2, 3, 5, 6\right]$,
- initial state (x_0): $\left[\text{-}0.5; \text{-}1\right]$,

4.4.1.2 Prediction Matrixes

From (4.32) and (4.33) we construct

$$\Phi = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

$$\Gamma = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

4.4.1.3 Time Constraints

$$c = \begin{bmatrix} \Delta t_{r_1} - \Delta t_{r_{1,min}} \\ \Delta t_{r_2} - \Delta t_{r_{2,min}} \\ \vdots \\ \Delta t_{r_{N-1}} - \Delta t_{r_{N-1,min}} \\ \Delta t_{r_N} - \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -2 + 1.625 \\ -1 + 0.635 \\ -2 + 1.625 \\ -1 + 0.625 \end{bmatrix} = \begin{bmatrix} -0.375 \\ -0.365 \\ -0.375 \\ -0.365 \end{bmatrix} \tag{4.53}$$

4.4.1.4 Results

Simulating this problem for 50 cycles we have

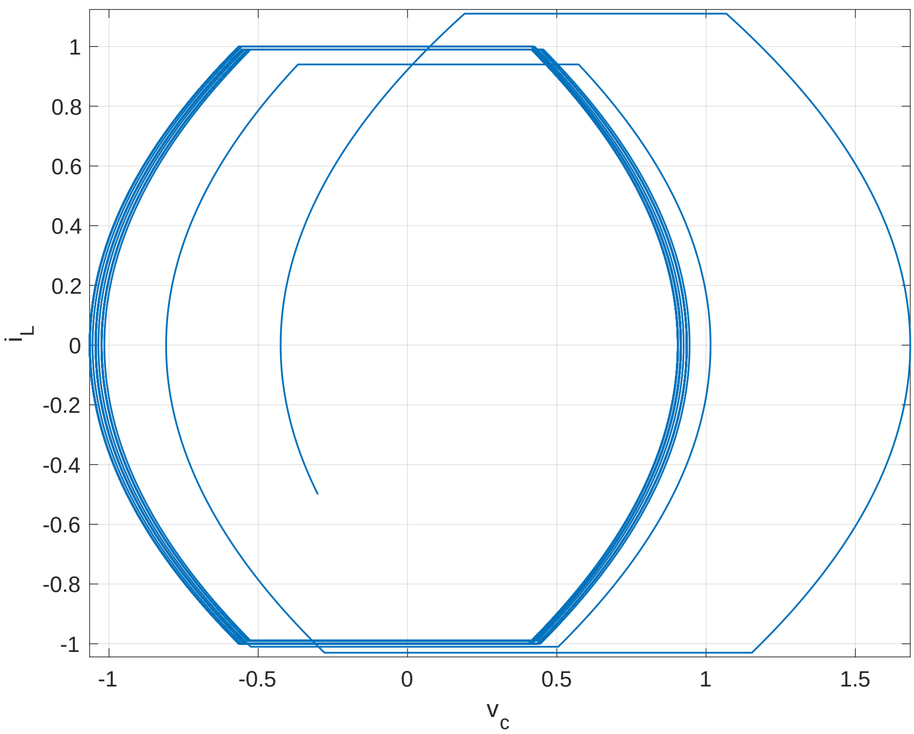

FIGURE 4.2 – State Trajectory

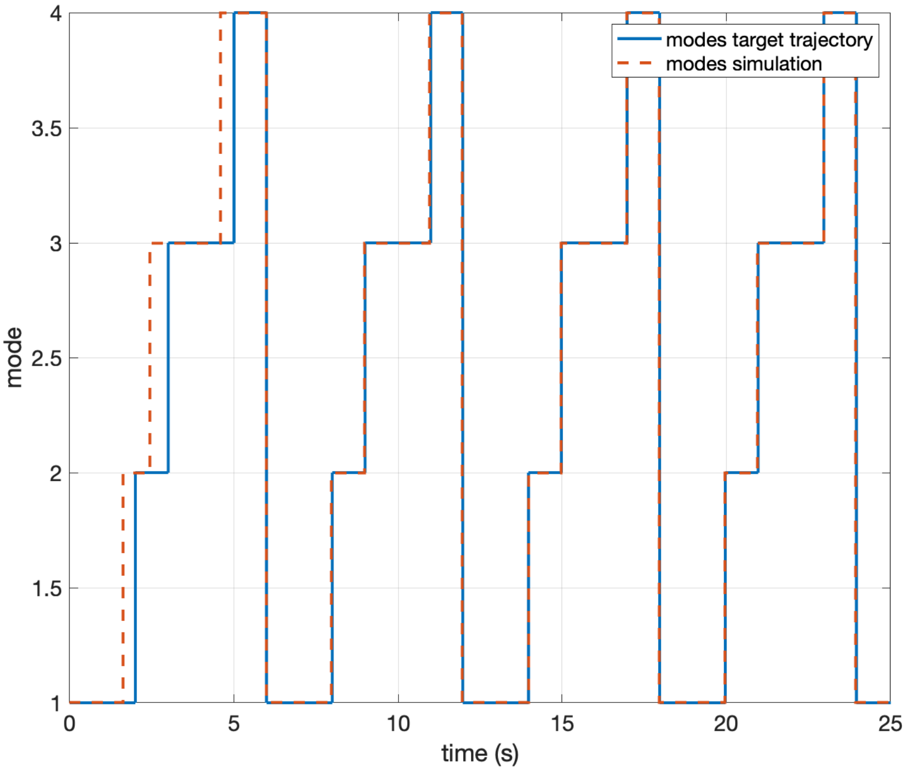

FIGURE 4.3 – Operation mode switch

4.4.2 Buck-boost converter

4.4.2.1 Parameters of the Simulation

- Ω: $\left[1, 2\right]$,
- s_{max} is set to 2,
- The dynamics of the system:

$$A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -1/(RC) \end{bmatrix} \quad A_2 = \begin{bmatrix} 0 & 1/L \\ -1/C & -1/(RC) \end{bmatrix}$$
$$b_1 = \begin{bmatrix} E/L \\ 0 \end{bmatrix} \qquad b_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{4.54}$$

where R = 1, L = 1, C = 1, E = 1,

- t_{step} is set to 1e-5.
- x_{ref} is set to $\left[2, -1\right]$.
- Q which is a diagonal matrix with both diagonal elements equal to 1.
- $T_{p_{max}}$ set to 1.
- T_s defined as $\left[0, 0.2513520, 0.5914520\right]$.
- x_0 set to $\left[1.870801, -1.119853\right]$.

4.4.2.2 Computation of the Target Trajetory

$$T_s = \begin{bmatrix} 0 & 0.2500021 & 0.5004299 \end{bmatrix} \tag{4.55}$$

4.4.2.3 Prediction Matrixes

$$\Phi = \begin{bmatrix} 1 & 0 \\ 0 & 0.7788 \end{bmatrix} \tag{4.56}$$

$$\Gamma = \begin{bmatrix} 1 \\ 1.1108 \end{bmatrix} \tag{4.57}$$

4.4.2.4 Time Constraints

$$c = \begin{bmatrix} \Delta t_{r_1} - \Delta t_{r_{1,min}} \\ \Delta t_{r_2} - \Delta t_{r_{2,min}} \\ \vdots \\ \Delta t_{r_{N-1}} - \Delta t_{r_{N-1,min}} \\ \Delta t_{r_N} - \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -0.25 + 0.25 \\ -0.2504 + 0.25 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.04 \end{bmatrix} \tag{4.58}$$

4.4.2.5 Results

Simulating 50 cycles:

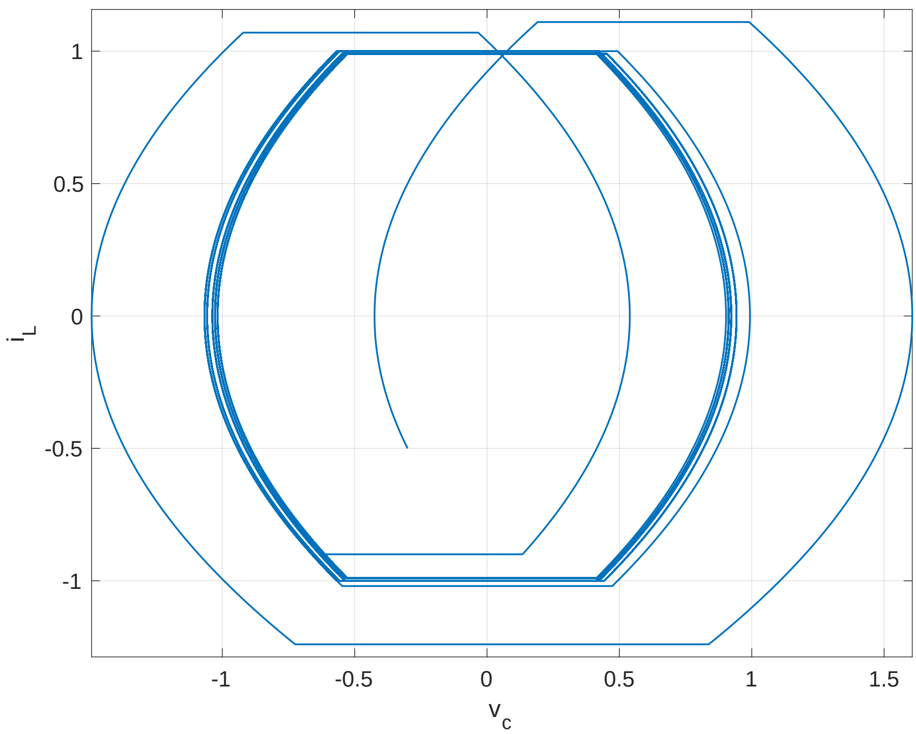

FIGURE 4.4 – fig 1

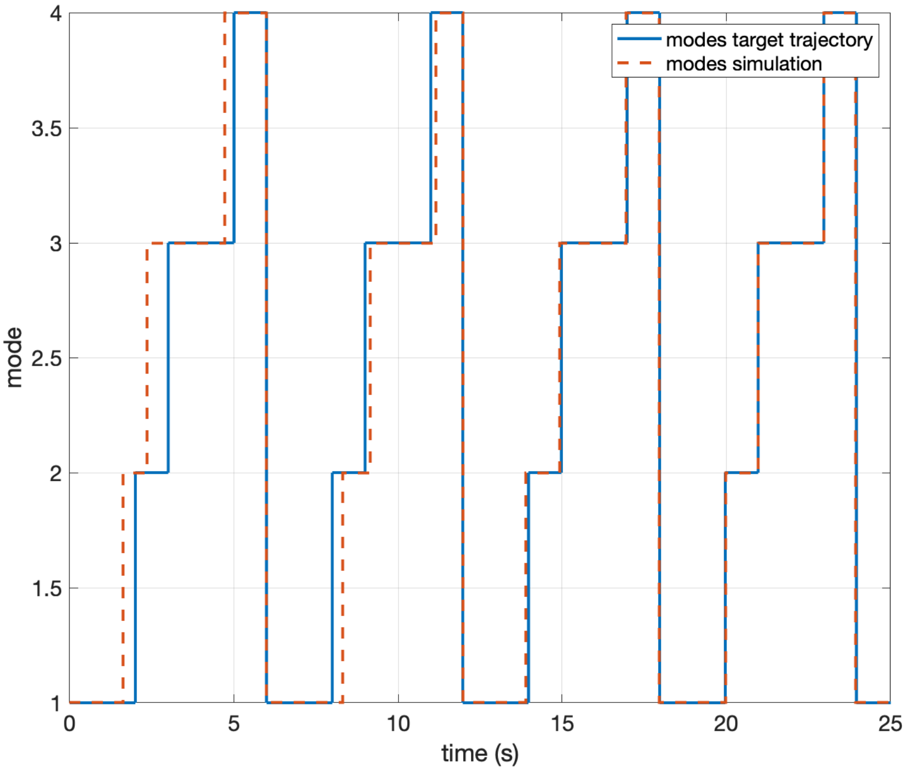

FIGURE 4.5 – fig 2

4.4.3 Patino 2

4.4.3.1 Parameters of the Simulation

The values used to simulate the Buck-boost converter example are:

- Ω: $\Big[1, 2, 4, 8, 3, 1, 5, 8, 5\Big]$

- s_{max} is set to 12.

- The dynamics of the system are defined by

$$A_i = \begin{bmatrix} 0 & 0 & \Big(u_2(i) - u_1(i)\Big)/C1 \\ 0 & 0 & \Big(u_3(i) - u_2(i)\Big)/C2 \\ \Big(u_1(i) - u_2(i)\Big)/L & \Big(u_2(i) - u_3(i)\Big)/L & -R/L \end{bmatrix}$$

$$b_i = \begin{bmatrix} 0 \\ 0 \\ \Big(E/L\Big)u_3(i) \end{bmatrix} \tag{4.59}$$

where R = 10.0 Ω, L = 10.0 mH, C_1 = 40.0 μF, C_2 = 40.0*1e-6 μF, E = 30.0 V and $u(i)$ is

TABLE 4.1 – heading

mode (i)	u_1	u_2	u_3
1	0	0	0
2	0	0	1
3	0	1	0
4	0	1	1
5	1	0	0
6	1	0	1
7	1	1	0
8	1	1	1

4.4.3.2 Computation of the Target Trajetory

$$T_s = \begin{bmatrix} 0 & 0.0639 & 0.0878 & 0.1119 & 0.1344 & 0.1567 & 0.2065 & 0.23 & 0.2529 & 0.2756 \end{bmatrix} \tag{4.60}$$

4.4.3.3 Prediction Matrixes

$$\Phi = \begin{bmatrix} 0.999174 & 0.001900 & 0.447676 \\ -0.002031 & 0.999915 & -0.000386 \\ -0.002208 & -0.000290 & 0.776298 \end{bmatrix} \tag{4.61}$$

$$\Gamma = \begin{bmatrix} -0.054681 & -2.586194 & -0.084235 \\ -2.480654 & 2.467091 & -0.078111 \\ 2.419040 & 0.052840 & -0.081930 \\ -2.481870 & 2.370819 & 0.179230 \\ 2.535489 & -2.594318 & 0.099365 \\ 2.651079 & 0 & -0.096018 \\ -2.464881 & 0 & -0.188814 \\ 0 & 0 & 0.201280 \end{bmatrix}^T * \times 10^4 \tag{4.62}$$

4.4.3.4 Time Constraints

$$c = \begin{bmatrix} \Delta t_{r_1} - \Delta t_{r_{1,min}} \\ \Delta t_{r_2} - \Delta t_{r_{2,min}} \\ \vdots \\ \Delta t_{r_{N-1}} - \Delta t_{r_{N-1,min}} \\ \Delta t_{r_N} - \Delta t_{r_{N,min}} \end{bmatrix} = \begin{bmatrix} -0.638811 + 0.20 \\ -0.239318 + 0.20 \\ -0.240440 + 0.20 \\ -0.225094 + 0.20 \\ -0.223339 + 0.20 \\ -0.497696 + 0.20 \\ -0.235714 + 0.20 \\ -0.228086 + 0.20 \\ -0.227575 + 0.20 \end{bmatrix} \times 10^{-4} = \begin{bmatrix} -0.438811 \\ -0.039318 \\ -0.040440 \\ -0.025094 \\ -0.023339 \\ -0.297696 \\ -0.035714 \\ -0.028086 \\ -0.027575 \end{bmatrix} \times 10^{-4} \tag{4.63}$$

4.4.3.5 Results

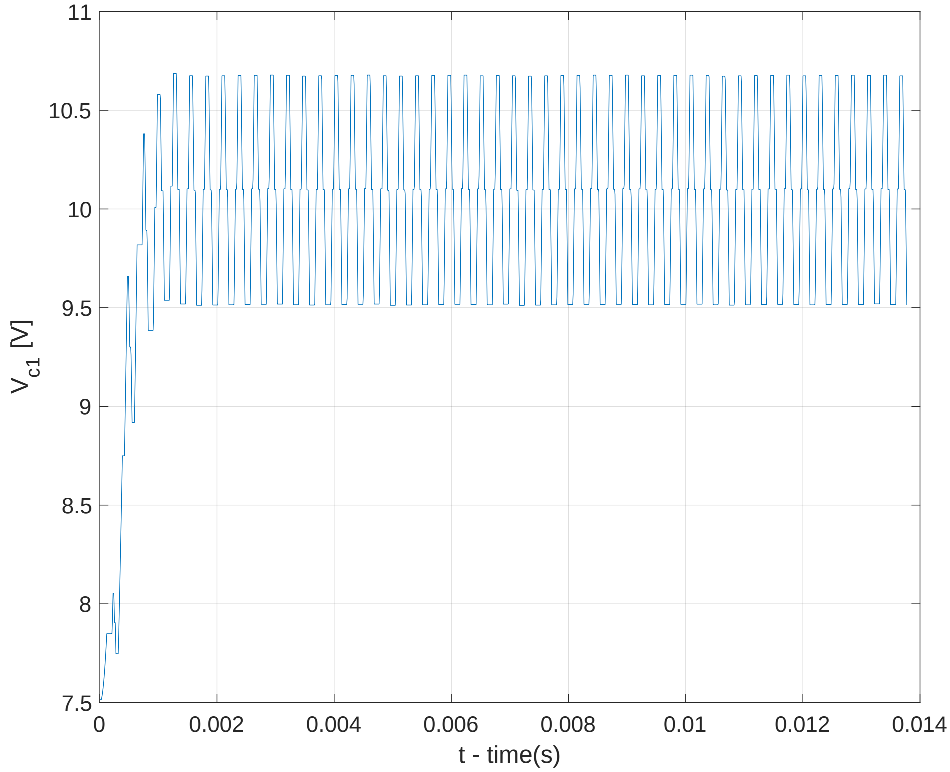

FIGURE 4.6 – fig 1

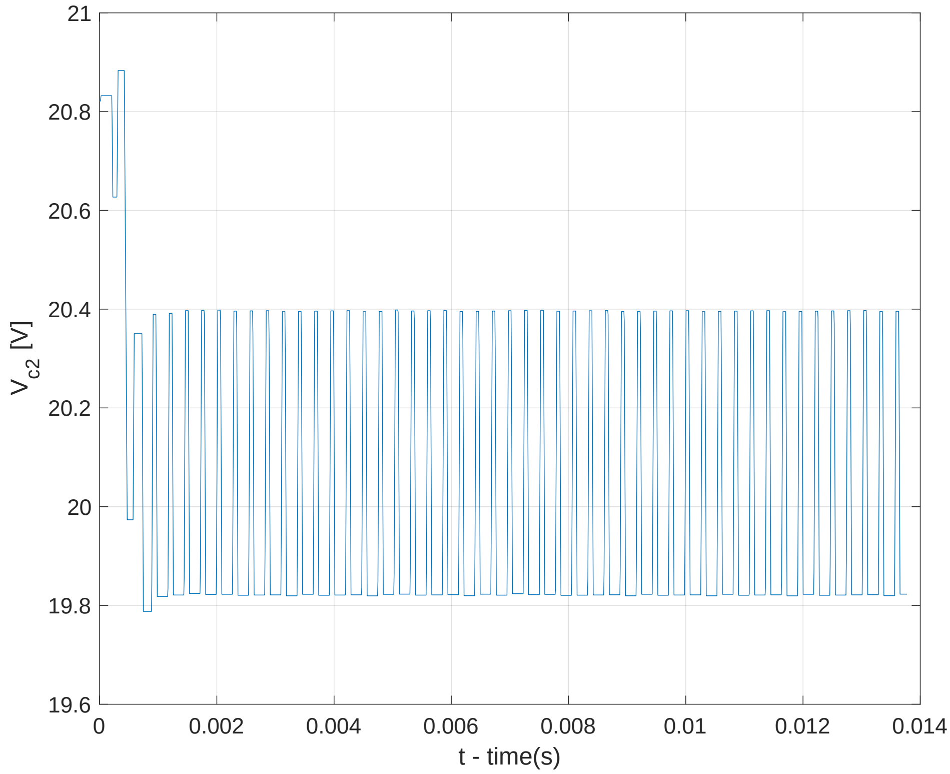

FIGURE 4.7 – fig 2

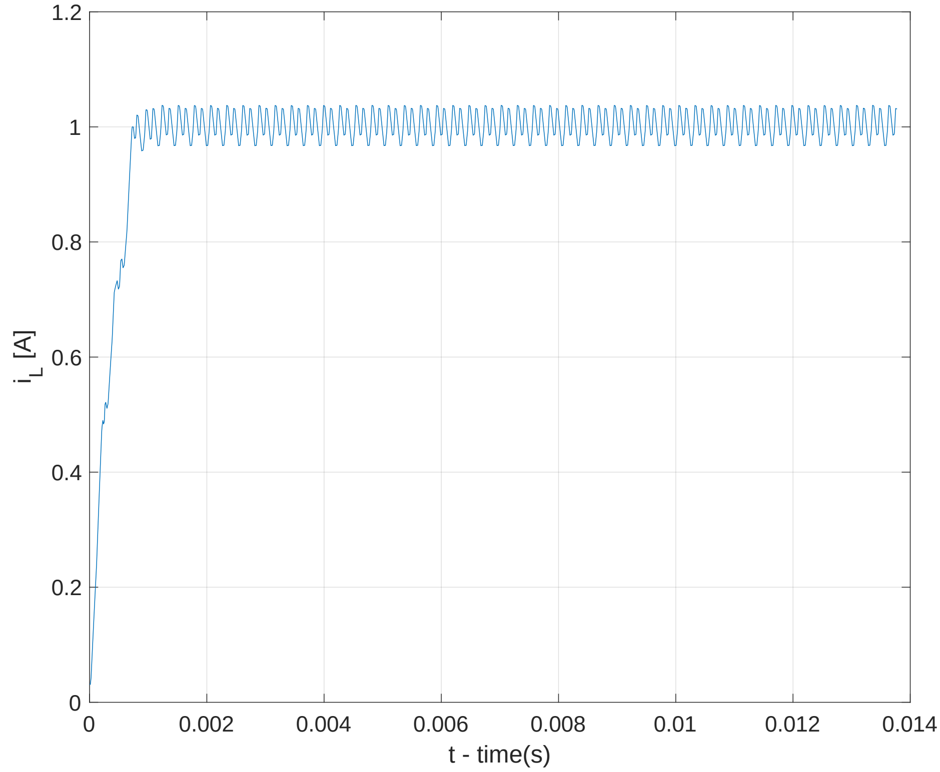

FIGURE 4.8 – fig 3

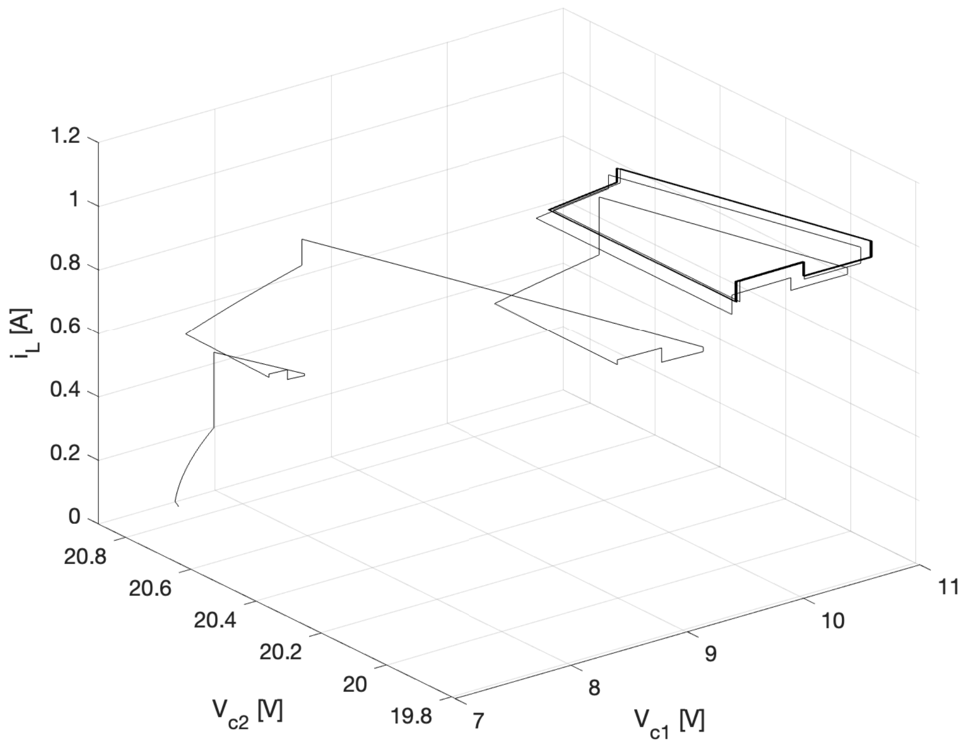

FIGURE 4.9 – fig 4

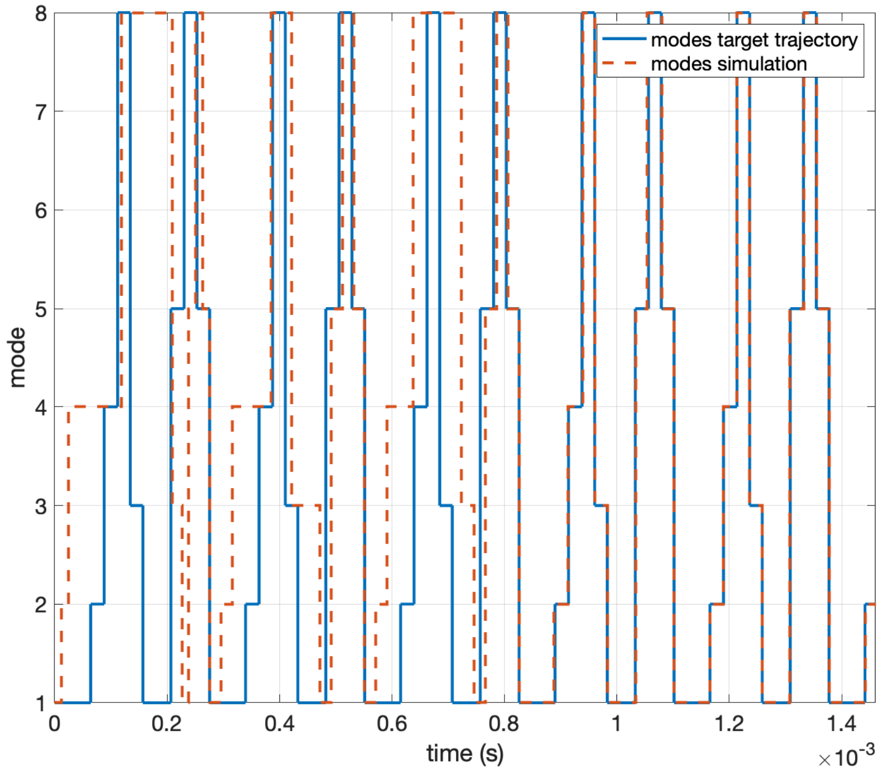

FIGURE 4.10 – Variant Dynamic Circuit 2: State Trajectory

4.5 Conclusion

mostrar o modelo em malha aberta indo para a sua trajetoria ciclica estavel junto com o modelo com controle ativo mostrando a convergencia mais rapida

5 Conclusão

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Appendix A - Tópicos de Dilema Linear

A.1 Modelo de simulação

```
|  +engine
|   |__ CORES.m
|   |__ fun_custo_patino.m
|   |__ get_config_sim_matheus.m
|   |__ get_config_sim_patino_1.m
|   |__ get_config_sim_patino_2.m
|   |__ get_config.m
|   |__ get_otmin_opt.m
|   |__ get_ts.m
|   |__ get_x0.m
|   |__ get_xr.m
|   |__ nonlincon_patino.m
|   |__ otmin.m
|   |__ sim_1.m
|   |__ sim_n.m
|__ +mpc
    |__ construcao_modelo_instantes.m
    |__ matrizes_ss_mpc_dualmode_switching.m
    |__ mpc_dualmode_switching.m
```

A.1.1 get_x0

funcao de calculo da condicao inicial

A.1.2 get_xf

Texto do modelo de simulação sendo colocado aqui.

Annex A - Exemplo de um Primeiro Anexo

A.1 Uma Seção do Primeiro Anexo

Algum texto na primeira seção do primeiro anexo.

FOLHA DE REGISTRO DO DOCUMENTO

1. CLASSIFICAÇÃO/TIPO	2. DATA	3. DOCUMENTO Nº	4. Nº DE PÁGINAS
TD	25 de março de 2015	DCTA/ITA/DM-018/2015	65

5. TÍTULO E SUBTÍTULO:

Robust control of linear systems with Switched Actuators Subjected to Dwell-Time Constraints

6. AUTOR(ES):

Daniel Vieira

7. INSTITUIÇÃO(ÕES)/ÓRGÃO(S) INTERNO(S)/DIVISÃO(ÕES):

Instituto Tecnológico de Aeronáutica – ITA

8. PALAVRAS-CHAVE SUGERIDAS PELO AUTOR:

Cupim; Cimento; Estruturas

9. PALAVRAS-CHAVE RESULTANTES DE INDEXAÇÃO:

Cupim; Dilema; Construção

10. APRESENTAÇÃO: (**X**) **Nacional** ( ) **Internacional**

ITA, São José dos Campos. Curso de Mestrado. Programa de Pós-Graduação em Engenharia Aeronáutica e Mecânica. Área de Sistemas Aeroespaciais e Mecatrônica. Orientador: Prof. Dr. Adalberto Santos Dupont. Coorientadora: Prof^{a}. Dr^{a}. Doralice Serra. Defesa em 05/03/2015. Publicada em 25/03/2015.

11. RESUMO:

Atuadores chaveados são difíceis de lidar por conta de algumas restrições características de sua natureza, porém eles são eficientes em inúmeras aplicações da engenharia, sendo aplicado, por exemplo, em sistemas de controle de atitude de veículos espaciais. Assim, saber utilizar desse tipo de atuador se faz importante e necessário. O principal problema a ser resolvido nessa tese é o projeto de um controlador que garanta a robustez de sistema com atuadores chaveados submetidos a restrições temporais que limitam a janela de resposta de seus comandos. Nessa abordagem, é feita uma busca de trajetória alvo de ciclo limite e, então, é calculado o erro dos estados e aplicada a técnica de controle preditivo para correção do erro e levar o systema à trajetória cíclica alvo.

12. GRAU DE SIGILO:

(**X**) **OSTENSIVO** ( ) **RESERVADO** ( ) **SECRETO**